

Family Structure and Youth Employment Instability: Evidence on Job Quality and Match Stability

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Abstract

This paper studies how family structure during adolescence is associated with early-career employment instability after conditioning on observed human capital. Using weekly labor market histories from the NLSY97, I compare workers who lived with a single parent at age 16 with those who lived two resident parent figures. Workers from single parent households experience higher post-school unemployment, driven primarily by more frequent separations from employment rather than slower job finding rates. Differences in accepted matches also depend on the path into a job: jobs entered from unemployment are shifted toward lower observed quality, whereas jobs reached through direct job-to-job transitions have similar observed quality but higher subsequent separation risk. I interpret these facts through a job-ladder model with heterogeneous wage quality, nonwage match value, and latent match stability. The model allows family structure to shift the wage-quality distribution of jobs accessible from unemployment and the stability composition of jobs accessible while employed. The estimated model reproduces the main transition and accepted-job moments, and attributes most of the aggregate employment-instability gap to reduced access to stable opportunities during employed search. Policy counterfactual exercises show that increasing unemployment resources do not close the separation gap, while expanding the overall quantity of on-the-job offers reduces it only indirectly. By contrast, expanding access specifically to stable opportunities is substantially more effective, closing the gap with a much smaller increase in total offer access. The results highlight the composition and durability of accessible jobs as important margins of early-career inequality.

Keywords: Labor Market Inequality; Family Structure; Youth Unemployment; Job Ladder; Match Stability

1 Introduction

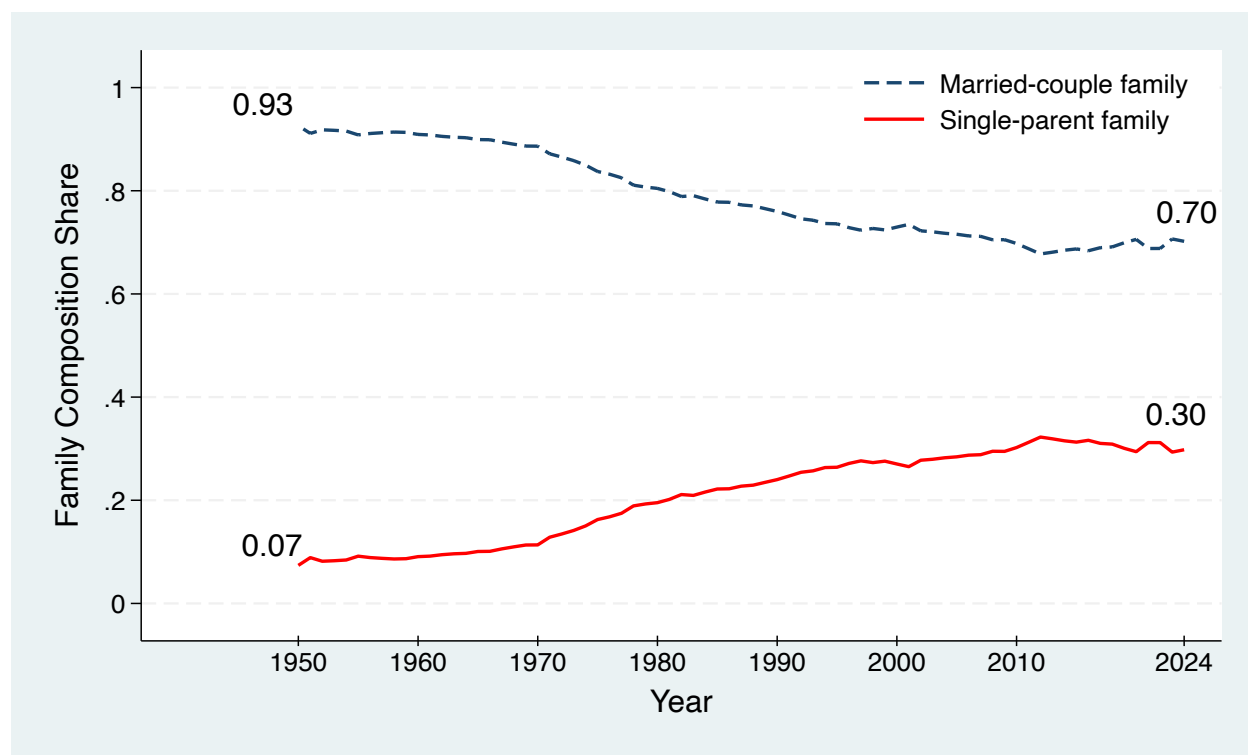
Research on labor market inequality among young workers often focuses on differences in employment, unemployment, and wages or earnings. These outcomes, however, summarize the cumulative effects of underlying labor market transitions. Educational attainment is an important source of heterogeneity in these outcomes, but even among workers with similar observed human capital, labor market trajectories may differ substantially in the frequency and characteristics of transitions across jobs and labor market states. Two workers with similar job-finding rates can experience very different employment trajectories if one repeatedly moves from employment into unemployment, enters less durable matches, or reaches lower quality jobs after an unemployment spell. A flow-based perspective is therefore particularly useful in early career, when employer changes are frequent and small differences in employment continuity can cumulate into persistent differences in experience and earnings.

This paper asks whether family structure during adolescence is systematically associated with these early-career ladder labor dynamics. Family structure is quantitatively important and relevant because family living arrangements have changed substantially in the United States. As shown in Figure 1, among families with own children under age 18, the single parent share increased from roughly 7 percent in 1950 to about 30 percent in 2024. While this secular change does not by itself imply a causal effect of family structure on adult outcomes, it makes important to understand whether family structure is associated with the way young adults enter jobs, move between employers, and experience unemployment.

Family structure may be correlated with many dimensions of the childhood environment and may itself shape human capital accumulation. Accordingly, the empirical analysis is designed to measure *conditional associations*: I ask whether family structure differences in labor market instability remain after adjusting for observed education and other background characteristics, and which labor market frictions account for those remaining differences.

I use the National Longitudinal Survey of Youth 1997 (NLSY97). I compare respondents who lived with a single parent at age 16 with those who lived with two resident parent figures. Family structure at age 16 is measured before the post-school labor market histories studied in the paper.

Figure 1: Family Composition among U.S. Families with Own Children under Age 18, 1950–2024



Notes: The figure reports the shares of married-couple (blue dashed line) and single-parent families (red solid line) among U.S. families with at least one own child under age 18. Single-parent families combine mother-only and father-only families. The denominator is the total number of families with own children under age 18, rather than all U.S. families or households. “Own children” include biological, step, and adopted children. The shares are calculated from the underlying family counts.

Source: Author’s calculations using the U.S. Census Bureau, Current Population Survey, March and Annual Social and Economic Supplements, Historical Families Table FM-1.

Family structure is defined from the household roster around at age 16. Throughout the analysis, I control for demographics, own educational attainment, parental education, and in richer specifications, measures of family and household resources.

Workers’ weekly employment histories allow me to distinguish unemployment stocks from the flows that generate them. I construct post-school labor market states at the weekly frequency, identify worker-employer spells and direct employer-to-employer (EE) moves, and measure the wage quality of accepted jobs using start-year-specific wage terciles. Note that, because the NLSY97 observes accepted jobs but not rejected offers, the empirical analysis is deliberately stated in terms of the composition of *accepted* worker-employer matches rather than the firm-side offer distribution.

The empirical findings are threefold. First, workers from single parent households have higher unemployment after school exit even after adjustment for observed human capital and family resources. The difference is not primarily driven by a job-finding rates. Unemployment-to-employment (UE) transition rates are similar across the two groups, whereas the employment-to-unemployment (EU) hazard is significantly higher for workers from single parent households.

Second, workers from single-parent households experience persistent disadvantages in job quality, beginning with their first jobs and continuing throughout their subsequent employment histories. To examine these differences, I compare the wage quality of accepted jobs across family structures. Workers from single-parent households are 2.11 percentage points more likely to enter the low-wage tercile and 1.87 percentage points less likely to enter the high-wage tercile in the fully controlled specification. I then distinguish jobs reached from unemployment from those reached through direct employment-to-employment (EE) transitions to identify where these differences emerge. The differences are substantially larger among reached from unemployment, whereas wage-quality differences among jobs reached through EE transitions are close to zero. Importantly, however, the less favorable wage composition of jobs reached from unemployment is not accompanied by a corresponding difference in their subsequent employment stability.

Third, family-structure differences in subsequent employment stability emerge primarily among jobs reached through direct EE transitions, rather than among jobs entered from unemployment. After accounting for observed wage quality and other covariates, workers from single-parent households have a 14.5% higher EU hazard than workers from two-parent households among jobs reached through direct EE transitions. In contrast, among jobs entered from unemployment, the difference in EU hazards is close to zero. This pattern contrasts with the wage-quality results: family-structure differences in accepted job quality are concentrated among jobs reached from unemployment, while differences in subsequent employment stability are concentrated among jobs reached through direct EE transitions.

In other words, employment instability that workers from single-parent households face when they find jobs is two-dimensional. When they find a job from unemployment status, the wage quality of job is less favorable on average. Once they get employed and try to change employers, the subsequent job is less durable, resulting in their falling back to unemployment status again.

This pattern implies that employment instability of workers from single-parent households is intertwined between the wage quality wedge and the stability wedge.

I use a quantitative job-ladder model to organize these facts and to separate two distinct margins of job access. The model builds on the heterogeneous match framework of Baydur and Mukoyama (2020). A match has three components: observed match quality q , nonwage match value x , and latent stability type s . Family structure enters the model only through differences in the *job offer distribution*. The first wedge, α_U , shifts the wage quality distribution of job offer accessible from unemployment. A negative α_U tilts the single parent distribution toward lower wage quality. The second wedge, γ_E , shifts the stability composition of job offer distribution during employed search. A negative γ_E lowers the probability that a job offer while on-the-job search to a worker from single parent household is of the stable type. I do not interpret either wedge as a literal firm-side offer distribution. Rather, each summarizes the opportunities that are effectively accessible to a worker after unmodeled information, network, geographic, and institutional frictions. The model does not require family-specific separation probabilities conditional on realized match type, family-specific preferences over nonwage amenities, or family-specific search effort.

A further feature of the final specification is persistence in wage quality across direct EE moves. In the NLSY97, 55.3 percent of direct EE moves with observed wage quality remain in the same wage tercile. An independent-draw job ladder substantially overpredicts upward wage quality mobility. I therefore allow the on-the-job search wage quality distribution to depend on the current wage-quality tercile through a common persistence parameter ρ_E . This parameter is common across family structure groups; the data do not provide sufficient evidence to justify a family-specific wage persistence wedge.

The preferred model fits the eleven core moments closely. In particular, it matches pooled UE and EE transitions, the EU rate of workers from two parent households, accepted job quality by entry labor state, the strong same-quality persistence of direct EE moves, and the higher quality-standardized EU risk of workers from single parent households among E-origin matches. The estimated unemployment-search quality wedge is negative, $\hat{\alpha}_U = -0.076$, and the on-the-job-search stability wedge is also negative, $\hat{\gamma}_E = -0.991$. With the baseline stable opportunity share normalized to one half, the latter implies a model-implied stable share of roughly 0.270 for *SP*

workers searching while employed. The model generates a stationary unemployment rate of 6.86 percent for two parent group and 7.89 percent for the single parent group.

I then implement three policy counterfactual exercises to reduce employment instability of workers from single-parent households: (i) an increase in unemployment income, (ii) an increase in job arrival frequency for unemployed workers, and (iii) an increase in job arrival frequency for employed workers. All three policy interventions do not change the underlying family-structure-related wedges but change only the relevant policy instruments. The first and second policy interventions make *SP* workers more selective in accepting higher-wage-quality jobs, so that the resulting composition of accepted wage-quality jobs improves. In addition, more frequent job arrival rates decrease the stationary unemployment rate of *SP* workers, but the first two policy interventions cannot close the overall EU hazard risks. The third policy also closes the stationary unemployment gap and even improves the wage quality of accepted jobs while unemployed. This is because more frequent job opportunities while employed increase the continuation value of employment overall. However, the underlying EU hazard gap still remains under all three policy interventions.

Related Literature This paper relates to two broad strands of literature: (i) research on family background, family structure, and intergenerational transmission, and (ii) research on young workers' labor market dynamics and early-career instability. I also draw on search and matching models to interpret the empirical findings through the lens of accepted job quality and match durability.

First, the paper relates to research on family background, family structure, and intergenerational transmission. A large literature studies how parental income, wealth, education, and household environment shape children's human capital accumulation and adult economic outcomes. Within this broad literature, work on family structure emphasizes that growing up with one parent rather than two may affect children through multiple channels, including economic resources, parental time, household stability, and access to informal support (McLanahan and Sandefur, 1994; Mayer, 1997; Kearney, 2023). Related work on parental resources shows that family wealth and transfers can shape early labor market search, reservation wages, unemployment

durations, and recovery from job loss (Mayer, 1997; Telemo and Visschers, 2022; Holmberg et al., 2024). Recent work also studies dynamic changes in family arrangements, such as divorce, and their consequences for children’s educational and adult outcomes (Johnston et al., 2025). I complement this literature by studying heterogeneity in the *dynamics* of young-adult employment, after conditioning on observed human capital and financial resources.

Second, this paper relates to research on young workers’ labor market dynamics and early-career instability. Young workers have high rates of mobility across jobs and labor market states, and early unemployment can have persistent consequences for later earnings and employment (Mroz and Savage, 2006; Kahn, 2010; Cajner et al., 2023). This literature establishes that the transition from school to work is a vulnerable stage of the life cycle. I add to this work by examining whether these dynamics differ systematically by family structure. This focus connects youth labor market instability to family background heterogeneity and shifts attention from job-finding rate alone and to the quality and durability of the matches young workers have.

To interpret these patterns, I estimate a quantitative model that builds on search and matching approaches with heterogeneous match characteristics. In particular, I follow the distinction in Baydur and Mukoyama (2020) between match quality and match stability, and adapt it to study heterogeneity by family structure. The structural contribution is a labor market transition model in which a predetermined family structure background characteristic indexes differences in the opportunity set relevant for job search.

Relative to these literature, this paper makes three contributions. First, it brings a flow-based perspective to the study of family structure and labor market outcomes. Rather than treating unemployment as a static outcome, the paper decomposes family-structure gaps into the transition rates that generate them. Second, it broadens the interpretation of family-background disadvantage beyond educational attainment and job-finding. The evidence points to accepted job quality and match durability as proximate margins through which early-life family structure is related to employment instability, even after adjusting for observed human capital. Third, it provides a structural decomposition of these margins. By separating accepted job quality differences from match stability differences, the model clarifies whether family-structure gaps operate mainly through the composition of jobs workers enter or through the durability of matches conditional

on job quality.

The remainder of the paper proceeds as follows. Section 2 describes the NLSY97 data, the construction of family structure at age 16, and the post-school weekly labor market panel. Section 3 documents reduced-form empirical evidence on labor market flows, accepted job quality, and origin-specific employment stability. Section 4 develops the quantitative job-ladder model. Section 5 describes calibration, estimation, discusses model fit, mechanism decompositions, and policy counterfactual exercises. Section 6 concludes.

2 Data and Measurement

2.1 NLSY97 and Family Structure at Age 16

I use data from the National Longitudinal Survey of Youth 1997 (NLSY97), a longitudinal survey that follows a nationally representative cohort of individuals born between 1980 and 1984 and first interviewed in 1997. Respondents were adolescents at the time of the initial interview and have been followed into adulthood. The NLSY97 is well suited for this paper for two reasons. First, it contains detailed information on respondents' family environments during adolescence, including household rosters that can be used to measure family structure around age 16. Second, it provides high frequency employment histories through event-history arrays, which allow labor market states and transitions to be measured at the weekly level.

The weekly structure of the employment histories is central to the analysis. Standard annual or point-in-time measure of employment status are useful for studying employment and unemployment levels, but they obscure the flows that generate those stocks. In contrast, the NLSY97 event-history arrays allow me to distinguish whether unemployment differences arise because some workers are less likely to find jobs, more likely to separate from jobs, or more likely to move in and out of the labor force. This distinction is particularly important for young workers, whose post-school employment histories involve frequent transitions across jobs and labor market states.

The main explanatory variable is family structure during adolescence. I construct this variable using household roster and relationship information from the NLSY97, which records the respondent's household members and their relationships to the respondent. The measure is based on

household composition around age 16 and classifies respondents according to whether they lived with one resident parent or with two resident parent figures. I code respondents with one resident parent and no second resident parent figure as having a single-parent background, and respondents with two resident parent figures as having a two-parent background. Throughout the paper, I refer to these groups as *SP* type parent and *TP* type workers.

This definition is designed to capture the family environment during late adolescence, before most respondents enter the post-school labor market. I focus on age 16 for two reasons. First, it is late enough to capture the family structure most relevant for the transition from school to work. Second, it measures family structure before respondents' own labor market histories, residential choices, or household formation decisions can directly shape the observed family environment. This timing allows family structure to be treated as a predetermined early-life condition relative to the post-school labor market transitions studied in the paper.

2.2 Post-School Labor Market States and Transitions

I construct weekly labor market states from the NLSY97 employment event-history arrays. For each respondent week, I classify the respondent as employed, unemployed, or not in the labor force.

The main analysis focuses on post-school weeks. For each respondent, I use schooling histories to identify the end of the schooling spell associated with their final educational attainment, and I retain weekly employment status observations after that point. This respondent-specific restriction ensures that the analysis captures labor market dynamics after school completion, rather than employment fluctuations during periods in which respondents are still primarily students. The resulting weekly panel follows respondents through their post-school employment histories and allows me to measure movements across employment, unemployment, and nonparticipation at high frequency.

Using the weekly panel, I construct labor market transitions between consecutive weeks. The main transitions are UE transitions and EU transitions. A UE transition occurs when a respondent is unemployed in week t and employed in week $t + 1$. An EU transition occurs when a respondent is employed in week t and unemployed in week $t + 1$. Weeks involving nonparticipa-

tion are retained to measure movements into and out of the labor force and to construct supplementary transition measures, but the main empirical analysis focuses on the distinction between job-finding from unemployment and separation from employment into unemployment.

2.3 Wage-Based Job Quality

To study the margins underlying employment instability, I augment the weekly employment panel with two sets of measures: characteristics of accepted jobs and worker-side resources. For the main analysis, I summarize the observed quality of an accepted match using its starting hourly wage and classify jobs into low, middle, and high wage quality categories relative to the distribution of starting wages in the job entry year. This wage-based measure provides a common observable dimension of job quality that maps directly into the quantitative model. The NLSY97 also reports a broader set of job characteristics, including occupation, industry, employer size, benefits, contract arrangements, hours, and work schedules. I use these in supplementary robustness exercises reported in the Appendix. The second set of measures captures worker-side resources and buffers and is described in the next subsection.

For each employed week, I identify the associated employer using the employer identifier recorded in the weekly employment history. Because job characteristics are generally reported at the employer or interview level rather than separately for every week, and because the same worker-employer match may appear in multiple interview records, I consolidate repeated records at the worker-employer-match level before linking the resulting match characteristics back to the weekly employment panel. The unit of observation in the accepted-job analysis is consequently an accepted worker-employer match rather than a respondent-week.

The main measure of observed job quality is the hourly wage at the beginning of the match. I classify each accepted job into the low, middle, or high tercile of the starting-wage distribution within the calendar year in which the match begins. Defining quality relative to the job-entry year removes changes in the nominal wage distribution over time and measures a job's position relative to other jobs begun in the same period.

The broader NLSY97 job information provides a complementary check on this wage-based measure. These data include occupation and industry categories, employer size, union or collective-

bargaining coverage, temporary-agency, on-call, and contract arrangements, hours and schedule characteristics, and employer-provided benefits. Rather than incorporating these dimensions into the main definition of job quality, I use them to test whether the family structure differences in employment stability are accounted for by other observable characteristics of accepted matches. Detailed variable construction and the corresponding robustness results are reported in Appendix [X].

Finally, because the NLSY97 records jobs that workers accept but does not observe rejected offers, the empirical objects in this section should be interpreted as characteristics of accepted worker-employer matches, not as direct measures of the underlying firm-side offer distribution. The quantitative model therefore uses the observed composition of accepted matches to discipline latent opportunity distributions jointly with workers' acceptance decisions.

2.4 Family Resources and Worker-Side Buffers

Having defined the observed characteristics of accepted matches, I next turn to worker-side resources and buffers that may also be associated with post-school employment stability. The main empirical specifications control for parental education and financial resources measured before or around labor-market entry. These controls are intended to assess whether the family-structure gaps documented in the paper remain after conditioning on observable differences in the economic resources available to workers from different family backgrounds.

The first set of financial-resource measures comes from the initial NLSY97 survey round and includes parent-reported household income, net worth, and poverty status. I include missing-value indicators for these measures so that respondents with incomplete resource information remain in the estimation sample. I also construct household income and poverty measures around the respondent's school exit. The school-exit measures capture the financial environment closer to the transition into the post-school labor market and complement the earlier measures of parental resources. The main specifications sequentially add these resource blocks to the demographic, own-education, and parental-education controls.

I use a broader set of worker-side measures in supplementary robustness and mechanism exercises. These include household assets and debt observed later in young adulthood, local geo-

graphic characteristics, residential mobility around school exit, and proximity to parents. Geographic measures include metropolitan status, urban residence, and region, while parental proximity distinguishes respondents living with or near a parent from those living farther away. These variables are not part of the baseline specification; instead, they are used to assess whether the main family-structure patterns are sensitive to broader differences in workers' spatial environments and potential access to family support. The corresponding results are reported in Appendix [X].

I interpret all of these covariate adjustments descriptively. Family financial resources, residential location, and proximity to parents may themselves be related to earlier family structure and, in some cases, to subsequent labor-market choices. Accordingly, changes in the estimated family structure coefficients after adding these controls should not be interpreted as causal mediation effects. Rather, the specifications show how much of the conditional family-structure gap remains after accounting for different sets of observed resources and constraints.

2.5 Analysis Sample

The analysis sample consists of respondent-week observations after school exit for individuals with valid information on family structure at age 16 and nonmissing labor market states. I exclude weeks before age 16 and weeks that cannot be assigned to employment, unemployment, or nonparticipation. I also exclude observations with missing or inconsistent calendar information that prevents the construction of weekly transitions. For transition analyses, I require valid labor market states in consecutive weeks.

The main sample pools respondents across survey rounds and follows them through the available weekly employment histories. Standard errors are clustered at the respondent level to account for serial correlation in weekly observations. The empirical analysis includes controls for sex, race, birth cohort, own education, parental education, calendar time, and other baseline characteristics, depending on the specification.

3 Empirical Analysis

This section documents how post-school labor market outcomes differ by family structure during adolescence. I begin with weekly labor market status outcomes, distinguishing employment, unemployment, and nonparticipation. I then examine whether the unemployment gap reflects differences in transitions into employment or transitions out of employment. Next, I study the quality of accepted jobs and how accepted job quality evolves over workers' post-school employment histories. Finally, I examine employment transitions by the quality of the job at origin, distinguishing the probability that the origin worker-employer relationship ends from the probability of moving directly to another employer conditional on such an exit. This final decomposition provides the quality-specific transition moments that discipline the structural model.

3.1 Empirical Strategy

The empirical analysis is not designed to estimate the total effect of family structure on young adult labor market outcomes. Family structure may affect labor employment through many channels, including educational attainment and other forms of human capital accumulation. Instead, the main empirical question is whether family structure is associated with labor market instability even after adjusting for observed human capital. I therefore include own educational attainment controls, including final grade completed and final degree category, in the main specifications. The resulting estimates should be interpreted as human capital adjusted family structure gaps rather than total family structure effects.

Throughout the empirical analysis, I use sequential covariate adjustments for two distinct purposes.¹ In the weekly status and transition hazard specifications, additional controls serve as sensitivity checks: they show whether the family-structure gaps remain after conditioning on observed human capital, parental education, and household resources. In the accepted match and worker buffer specifications, the added covariates serve as descriptive accounting controls: they show whether the higher EU hazard is attenuated after conditioning on observed match condi-

¹Sequential covariate adjustments can be sensitive to the order in which correlated controls are added. Gelbach (2016) provides a formal omitted variable bias based decomposition for attributing coefficient movements to added covariates. I do not implement that decomposition here, so the sequential specifications should be interpreted descriptively rather than as an order-invariant decomposition.

Table 1: Post-School Weekly Labor Market Status by Family Structure

	Two-parent background	Single-parent background
Respondents	5,055	2,583
Post-school respondent-weeks	3,851,002	2,080,489
Employed weeks	3,114,884	1,477,860
Unemployed weeks	172,727	155,118
NILF weeks	563,391	447,511
Employment share	80.89%	71.03%
Unemployment share	4.49%	7.46%
NILF share	14.63%	21.51%
Unemployment rate among labor force weeks, ($U/(E + U)$)	5.25%	9.50%

Notes: This table reports post-school respondent weeks by family structure at age 16. Respondents are counted once within each family-structure group. Shares are computed over pooled respondent-week observations within each family-structure group. The unemployment share uses all post-school weeks as the denominator, while $U/(E + U)$ restricts the denominator to labor force weeks. NILF denotes not-in- the-labor-force.

tions and worker-side resources. I do not interpret these specifications as causal mediation estimates. Instead, they describe how the estimated family structure gaps change when conditioning on pre-specified, substantively motivated covariate blocks.

3.2 Post-School Weekly Labor Market Status

Table 1 first reports raw post-school weekly labor market status by family structure at age 16. Individuals from single-parent households have lower employment shares, higher unemployment, and nonparticipation shares during the post-school period. Among respondents from two-parent households, employed weeks account for 80.9% of all post-school weeks, while unemployed weeks and weeks out of the labor force account for 4.5% unemployed and 14.6%, respectively. Among respondents from single-parent households, the corresponding shares are 71.0%, 7.5%, and 21.5%. Restricting attention to labor force weeks, the unemployment rate is 5.3% for workers from two-parent households and 9.5% for workers from single-parent households.

To assess whether these differences remain after conditioning on observed characteristics, I estimate separate binary logit models for each weekly labor market status outcome and report ad-

justed probability differences between workers from single-parent and two-parent households.² For the binary family structure indicator, the reported difference is computed by changing the indicator from two-parent to single-parent households for each observation, holding all other covariates at their observed values, and averaging the resulting change in predicted probability over the estimation sample. I therefore refer to this quantity as an *adjusted probability difference*, rather than interpreting it as the effect of a marginal increase in single parenthood.

The dependent variables indicate whether a respondent is employed, unemployed, or out of the labor force in a given post-school week. I also estimate a binary logit model for unemployment conditional on labor force participation and an individual level regression of each respondent's post-school unemployment share, $U_i / (E_i + U_i)$. The respondent week specifications control for sex, race, birth year, birth month, final grade completed, final degree category, and calendar-quarter fixed effects. I then add parental education and first-round parental financial resources, including parental income, net worth, poverty status, and corresponding missingness indicators. Standard errors in the respondent-week specifications are clustered at the respondent level. The individual level unemployment share regression gives equal weight to each respondent.

Table 2 reports the most fully adjusted specifications, including demographics, own education, parental education, first-round parental resources, and calendar-quarter fixed effects. Conditional on these observed characteristics, the adjusted probability of employment is 2.0 percentage points lower for workers from single-parent households, while the adjusted probability of unemployment is 0.8 percentage points higher. Among labor force weeks, the adjusted unemployment probability is 1.0 percentage point higher for workers from single-parent households. This gap is equivalent to about 16% of the standardized two-parent benchmark of 6.1%. The NILF difference is positive but smaller and only marginally statistically significant. The individual level unemployment share regression also yields a positive and statistically significant difference, showing that the result is robust to giving each respondent equal weight rather than weighting individuals by the number of observed labor force weeks.

Taken together, these results in Table 2 point to unemployment as a key dimension of the

²A multinomial logit model is a natural alternative because the three weekly states are mutually exclusive. I use separate binary logits because they allow the adjusted probability difference between the two groups to be reported directly for each labor market state in percentage point terms.

Table 2: Adjusted Differences in Post-School Weekly Labor Market Status

	All post-school weeks			Labor force weeks	Individual
	Employed	Unemployed	NILF	Unemployed	$U_i / (E_i + U_i)$
SP-TP difference (% points)	-2.04*** (0.72)	0.76*** (0.23)	1.21* (0.64)	1.00*** (0.30)	1.44*** (0.38)
Standardized two-parent benchmark (%)	79.20	5.18	15.61	6.15	7.15
Observations	5,746,202	5,746,149	5,746,202	4,821,227	7,347
Respondents	7,347	7,347	7,347	7,347	7,347
Demographic controls	Yes	Yes	Yes	Yes	Yes
Own education controls	Yes	Yes	Yes	Yes	Yes
Parental education controls	Yes	Yes	Yes	Yes	Yes
First-round parental resources	Yes	Yes	Yes	Yes	Yes
Calendar-quarter fixed effects	Yes	Yes	Yes	Yes	No

Notes: Columns (1)–(4) report adjusted probability differences between workers from single-parent and two-parent households from binary logit models. The reported estimate is the average discrete change in predicted probability with the single-parent household indicator is changed from zero to one, holding all other covariates at their observed values. Columns (1)–(3) use all post-school respondent weeks, while Column (4) restricts the sample to labor force weeks. Column (5) reports the coefficient from an individual level OLS regression of each respondent’s post-school unemployment share, $U_i / (E_i + U_i)$. The workers from two-parent households are the reference group. The row “Standardized two-parent benchmark” reports the average predicted outcome obtained by setting the single-parent household indicator to zero for every observation in the corresponding estimation sample while retaining all other covariates at their observed values. Standard errors are reported in parentheses and are clustered at the respondent level in Columns (1)–(4) and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

post-school employment instability associated with family structure. Table 3 therefore focuses on unemployment among labor force weeks and examines how this gap changes as additional covariates are introduced.

In Table 3, the first specification conditions on demographics, own education, and calendar-quarter fixed effects. I then sequentially add parental education and two sets of family financial resource measures. First-round resources capture earlier household financial circumstances using parental-reported income, net worth, and poverty status from the initial NLSY97 survey, whereas school-exit resources measure household income and poverty closer to the respondent’s transition out of schooling. The latter provide a more proximate measure of family financial circumstances around labor market entry and allow us to assess whether the unemployment difference is sensitive to changes in household resources between the initial survey and school exit period.

The adjusted unemployment difference is 1.35 percentage points in the baseline specification and changes only slightly, to 1.28 percentage points, after adding parental education. Adding first-round parental resources reduces the difference to 1.00 percentage point. When resources mea-

Table 3: Labor Force Unemployment Gap and Family-Background Controls

	Probability of Unemployment among labor force weeks			
	Baseline	+ Parent edu.	+ R1 resources	+ Exit resources
SP-TP difference (% points)	1.35*** (0.30)	1.28*** (0.30)	1.00*** (0.30)	0.61** (0.29)
Standardized two-parent benchmark (%)	6.08	6.10	6.15	6.31
Observations	4,824,527	4,824,527	4,821,227	4,821,227
Respondents	7,406	7,406	7,347	7,347
Demographic controls	Yes	Yes	Yes	Yes
Calendar-quarter fixed effects	Yes	Yes	Yes	Yes
Birth year/month controls	Yes	Yes	Yes	Yes
Own education controls	Yes	Yes	Yes	Yes
Parental education controls	No	Yes	Yes	Yes
First-round parental resources	No	No	Yes	Yes
School-exit resources	No	No	No	Yes

Notes: The outcome is an indicator for unemployment among labor force respondent weeks. The table reports adjusted probability differences between workers from single-parent and two-parent households from logit models. For each specification, the reported estimate is the average discrete change in predicted unemployment probability when the single-parent household indicator is changed from zero to one, holding all other covariates at their observed values. The “Standardized two-parent benchmark” is the average predicted unemployment probability obtained by setting the single-parent household indicator to zero for all observations. Standard errors are clustered at the respondent level and reported in parentheses. . * * * $p < 0.01$, * * $p < 0.05$, * $p < 0.10$.

sured around school exit are also included, the difference falls further to 0.61 percentage points and remains statistically significant at the 5 percent level. Thus, the estimated difference becomes smaller as observed family financial resources are added, particularly when conditioning on resources measured closer to labor market entry. Because these resources may themselves be related to earlier family structure and subsequent labor market choices, I interpret the sequential changes descriptively rather than as causal mediation. The results are also robust to alternative controls for age and to estimating the unemployment specification using a linear probability model. These robustness checks are reported in Appendix [X].

One remaining concern is that the association between family structure and unemployment that remains after conditioning on education and financial resources may reflect persistent worker characteristics not captured by these controls, including noncognitive traits. This possibility is motivated by the skill-formation literature, which emphasizes that family environments can shape persistent cognitive and noncognitive abilities that matter for later socioeconomic outcomes (Cunha et al., 2006). Appendix B examines this possibility using the noncognitive measures available in

the NLSY97: Goldberg-based measures of conscientiousness and agreeableness, together with an earlier measure of behavioral and emotional problems.

These observed noncognitive characteristics are predictive of unemployment, with higher conscientiousness in particular associated with a lower unemployment probability. However, adding these measures reduces the estimated association between family structure and unemployment by only about 9–16 percent in same-sample comparisons. This relatively small amount of attenuation suggests that noncognitive characteristics may account for part of the remaining association, but it is not the primary underlying mechanism behind the higher unemployment rate of workers from single-parent households. Appendix B describes the detailed results and discusses the limitations of this interpretation.

3.3 UE and EU Transition Hazards

The weekly status results show that workers from single-parent households have higher unemployment rates, but unemployment is a stock variable. A higher unemployment rate can arise because workers enter into unemployment more often, because they exit from unemployment more slowly, or both. I therefore decompose the unemployment gap into weekly transition hazards.

I estimate discrete-time transition hazard models for two transitions by using a complementary log-log specification.³ The first transition is the unemployment-to-employment transition, UE, where the risk set consists of unemployed respondent weeks and the outcome equals 1 if the worker is employed in the following week. The second transition is the employment-to-unemployment transition, EU, where the risk set consists of employed respondent weeks and the outcome equals 1 if the worker is unemployed in the following week. The empirical specification is

$$\log[-\log(1 - h_{it})] = \alpha + \beta \text{SingleParent}_i + X_i' \gamma + \lambda_t + \delta \log D_{it}$$

where h_{it} is the conditional probability that respondent i makes the relevant transition in week t ,

³Because labor market transitions are observed only at weekly intervals rather than at exact dates, spell durations are grouped in discrete time. I therefore estimate discrete-time hazard models using a complementary log-log link, which accommodates the weekly observation structure while retaining a proportional hazards interpretation.

Table 4: Weekly Transition Hazards by Family Structure

	UE Transition			EU Transition		
	Main	+ R1 resources	+ R1 and exit resources	Main	+ R1 resources	+ R1 and exit resources
<i>SP-to-TP</i> hazard ratio	0.9647 (0.0339)	0.9861 (0.0361)	1.0072 (0.0369)	1.1481*** (0.0400)	1.1262*** (0.0395)	1.1048*** (0.0398)
Observations	325,146	321,635	321,635	4,539,960	4,539,960	4,539,960
Transition events	7,533	7,533	7,533	8,842	8,842	8,842
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes
Own education controls	Yes	Yes	Yes	Yes	Yes	Yes
Parental education controls	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-time controls	Yes	Yes	Yes	Yes	Yes	Yes
Duration controls	Yes	Yes	Yes	Yes	Yes	Yes
First-round parental resources	No	Yes	Yes	No	Yes	Yes
School-exit resources	No	No	Yes	No	No	Yes

Notes: This table reports exponentiated coefficients from discrete-time complementary log-log transition-hazard models. Columns (1)–(3) estimate UE transitions on unemployed respondent-weeks. Columns (4)–(6) estimate EU transitions on employed respondent weeks. The coefficient on single-parent household is reported as a hazard ratio. A hazard ratio above one indicates a higher weekly transition hazard relative to workers from two-parent households. Standard errors are clustered at the respondent level and reported in parentheses. Demographic controls include sex, race, and birth cohort. Own education controls include completed schooling measures. First-round parental resources include parental income, net worth, poverty status, and missingness indicators. School-exit resources include income and poverty measures around school exit and missingness indicators. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

given that the respondent is in the corresponding risk set at the beginning of the week. SingleParent_i is an indicator for living with a single parent at age 16, X_i includes demographic and education controls, λ_t denotes calendar-time controls, and D_{it} is the number of weeks elapsed in the current spell. In extended specifications, I add parental education and household resource controls. Standard errors are clustered at the respondent level. I report exponential coefficient, $\exp(\beta)$, as hazard ratios.

Table 4 shows that the unemployment gap is driven primarily by higher transitions from employment into unemployment rather than by slower transitions from unemployment into employment. The estimated UE hazard ratio between *SP* workers to *TP* workers is close to 1 and statistically insignificant across specifications. This result implies that UE risk of workers is not basically different across two family-structure groups. In contrast, the EU hazard ratio is 1.148 in the main specification, which implies that workers from single-parent households face roughly a 15 percent higher weekly risk of moving from employment to unemployment. Adding first-round parental resources reduces the hazard ratio to 1.126, and adding both first-round and school-exit resources reduces it further to 1.105. The estimate remains above one and statistically significant throughout.

Put together, these transition results indicate that the higher unemployment associated with single-parent household is not primarily explained by a slower reemployment transition once unemployment has begun. Instead, the main difference appears on the employment side of the labor market: workers from single-parent households are more likely to move from employment into unemployment.

The fact that the EU difference remains after conditioning on observed human capital and family resources raises the next question: does the employment instability associated with single-parent household reflect sorting into different types of accepted jobs, or greater fragility even among jobs of similar observed quality? I examine these margins in two steps. I first study the wage-quality composition of accepted jobs. I then distinguish jobs by whether they are entered from unemployment or from employment, since job search behavior and the opportunities available to workers can differ substantially across these two labor market states (Holzer, 1987; Blau and Robins, 1990; Faberman et al., 2022). I then examine whether family-structure differences in subsequent separation risk are still present after conditioning on observed job quality, separately for jobs entered from unemployment and from employment.

An alternative explanation is that employment stability depends partly on parental proximity and informal family support. First, parental proximity may affect employment stability in either direction. Nearby parents may provide transportation, emergency assistance, or other forms of practical support, but living close to parents may also reflect family-care obligations or geographic attachment that restrict workers' mobility and increase demands on their time. Second, grandparent availability provides a complementary test of the support channel: an additional family member may supply time or logistical assistance that partly offsets differences in the worker-side buffers associated with single-parent household.

Appendix C examines these possibilities using measures of parental proximity and grandparent availability. Because these measures do not directly observe actual transfers of time, transportation, care, or other forms of assistance, and may themselves reflect earlier family circumstances and endogenous location or co-residence choices, I interpret the results as descriptive mechanism diagnostic rather than as causal mediation estimates.

3.4 Accepted Job Quality

3.4.1 Average Differences in Accepted Job Quality

Having established that the unemployment gap by family structure is driven primarily by a higher EU hazard rather than by a lower job-finding rate, I next examine whether workers from different family structure enter systematically different types of accepted jobs. I use the hourly wage at the beginning of each worker-employer match as an observable proxy for accepted-job quality.⁴ Specifically, I classify each accepted job into the low, middle, or high tercile of hourly wage distribution. Thus, throughout this subsection, job quality refers to a wage-based measure of the quality of accepted jobs.

I estimate multinomial logit models of accepted-job quality and report average discrete change associated with single-parent household in the probability of entering the low and high wage terciles. The specifications sequentially add job-entry-year and accepted-job-order fixed effects, demographic characteristics, parental education, and the worker's own educational attainment.

Table 5 reports the results. In the unadjusted specification, workers from single-parent households are 9.1 percentage points more likely to enter the low wage tercile and 10.5 percentage points less likely to enter the high wage tercile. These differences change little after adding job-entry-year and accepted-job-order fixed effects, but become progressively smaller as demographic characteristics, parental education, and the worker's own education are added.

In the full specification, single-parent household is associated with a 2.1 percentage point higher probability of entering the low wage tercile and a 1.9 percentage point lower probability of entering the high wage tercile. The former estimate is statistically significant at the 5 percent level, while the latter is significant at the 10 percent level. Thus, much of the raw difference in accepted-job quality is associated with observed worker characteristics and family resources, but a modest difference in the composition of accepted jobs remains after conditioning on these observables.

⁴The wage-based measure captures one observable dimension of job quality and should not be interpreted as a comprehensive measure of nonwage amenities, working conditions, or job security.

Table 5: Family Structure and Accepted Job Quality

	<i>SP-TP</i> predicted probability difference				
	(1)	(2)	(3)	(4)	(5)
Low-quality job (% points)	9.08*** (1.03)	8.91*** (1.03)	6.05*** (1.03)	4.60*** (1.04)	2.11** (0.98)
High-quality job (% points)	-10.48*** (1.07)	-10.31*** (1.07)	-7.08*** (1.10)	-4.69*** (1.09)	-1.87* (1.05)
Entry-year fixed effects	No	Yes	Yes	Yes	Yes
Accepted-job-order fixed effects	No	Yes	Yes	Yes	Yes
Demographic controls	No	No	Yes	Yes	Yes
Parental-education controls	No	No	No	Yes	Yes
Own-education controls	No	No	No	No	Yes
Accepted-job observations	34,482	34,482	34,482	34,482	34,113

Notes: The table reports average marginal effects from multinomial logit models of accepted-job quality. Job quality is measured using terciles of start-year-adjusted hourly wages, with the middle tercile serving as the base outcome. The reported coefficients are the average discrete change in percentage points, in the probability of entering the indicated quality category associated with single parent household, relative to two parent household. The unit of observation is an accepted worker-employer match.

Accepted-job-order fixed effects distinguish job orders one through ten and pool orders eleven and above. Demographic controls include sex, race, and birth-year indicators. Parental-education controls include categorical measures of maternal and paternal educational attainment. Own-education controls include final grade completed and final educational-attainment category. Standard errors, clustered at the worker level, are reported in parentheses.

The estimates represent conditional associations. The attenuation across specification should not be interpreted as a causal mediation decomposition. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.4.2 Accepted Job Quality over Employment Histories

The previous pooled estimates show that workers from single-parent households enter lower quality jobs on average. I next examine whether this job-quality disadvantage is confined to initial labor market entry or recurs as workers move through subsequent jobs. A disadvantage that reappears after the first job would indicate that the accepted-job quality gap is a recurring feature of the employment history, rather than a one-time condition at school exit. I therefore compare accepted-job quality at the first and subsequent post-school jobs and then distinguish subsequent jobs by the worker's labor market state immediately before job entry.

Table 6 examines where the accepted job quality gap arises over post-school employment histories. Panel A shows that the accepted-job quality disadvantage is not confined to the first post-school job. At the first accepted job, single-parent household is associated with a 4.82 percentage point lower probability of entering the high wage tercile, while the corresponding difference in

the low wage tercile is positive but statistically insignificant. Among subsequent jobs, the pattern shifts toward the lower tail of the wage distribution: single-parent household is associated with a 2.17 percentage point higher probability of entering the low wage tercile, while the difference in the high wage tercile is smaller and statistically insignificant. Thus, the disadvantage reappears after the initial school-to-work transition, although the margin along which it appears changes over the employment history.

Panel B then distinguishes subsequent accepted jobs by the worker's labor market state immediately before job entry. The largest differences appear among jobs entered from unemployment. Conditional on the full set of controls, workers from single-parent households are 5.70 percentage points more likely to enter the low wage tercile following unemployment and 4.25 percentage points less likely to enter the high wage tercile. By contrast, there is little evidence of a comparable difference among jobs entered directly from employment: the estimated differences in both the low and high wage terciles are close to zero. The estimates for jobs entered from nonparticipation are also smaller and statistically insignificant.

This result sharpens the interpretation of the transition results in Table 4. Workers from single-parent households do not have substantially lower UE transition rates, but conditional on reemployment, the *composition* of accepted jobs is less favorable. The difference is therefore not primarily in whether workers leave unemployment, but in the types of jobs they enter when they do.

These results establish that the accepted-job quality disadvantage is recurring and is especially pronounced for jobs entered from unemployed. They do not, however, show whether workers from different family structure subsequently experience different employment stability *within* jobs of similar observed quality. I examine this margin in the next subsection.

3.5 Separating Job-Quality Composition from Match Stability

The previous subsection establishes a family-structure difference in the *composition* of accepted jobs: workers from single-parent households are more likely to enter low quality jobs, particularly following unemployment. Because lower wage jobs tend to be associated with higher separation risk (Jarosch, 2023), this composition difference could itself contribute to the higher EU hazard

Table 6: Accepted Job Quality over Post-School Employment Histories

	<i>SP-TP</i> predicted probability difference		Sample size	
	Low tercile (% points)	High tercile (% points)	Jobs	Workers
<i>Panel A. First and subsequent accepted jobs</i>				
First post-school accepted job	1.32 (1.44)	−4.82*** (1.70)	3,965	3,965
Subsequent accepted jobs	2.17** (1.06)	−1.48 (1.11)	30,148	5,660
<i>Panel B. Subsequent accepted jobs by search state</i>				
Employment	0.72 (1.31)	0.07 (1.58)	10,920	3,968
Unemployment	5.70*** (1.87)	−4.25** (1.78)	4,825	2,465
Not-in-the-labor-force	1.88 (1.32)	−1.06 (1.24)	13,661	4,470

Notes: The table reports average marginal effects, in percentage points, of a single parent household on the probability that an accepted job falls in the low or high tercile of the starting-wage distribution. The middle wage tercile is the omitted outcome. Wage terciles are defined relative to the calendar year in which the job begins. The unit of observation is an accepted worker-employer match.

Panel A estimates separate multinomial logit models for the first post-school accepted job and for subsequent accepted jobs.

Panel B estimates separate multinomial logit models according to the worker's labor market state immediately before entry into the subsequent accepted job. Employment, unemployment, and not-in-the-labor-force denote the three reported origin states. Subsequent jobs with another or unclassified origin state are not reported in Panel B.

All specifications include job-entry-year fixed effects, sex, birth-year fixed effects, race, maternal and paternal education, final grade completed, and final educational-attainment category. Standard errors are clustered by worker and reported in parentheses. Workers with multiple subsequent accepted jobs contribute multiple observations. All accepted jobs begin after the respondent's terminal observed schooling period.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

documented in Table 4. The relevant question is therefore whether the family-structure difference in subsequent employment stability remains once matches are evaluated under a common distribution of observed job quality.

I next examine whether there is an additional stability channel beyond this sorting margin. To separate job-quality composition from subsequent match stability, I classify each worker-employer match by its entry path. I distinguish jobs reached through a direct job-to-job transition from those entered from unemployment. I then estimate a weekly complementary log-log model of the subsequent EU transition that fully interacts family structure, entry path, and the wage quality tercile of the accepted match. The specification includes demographic characteristics, own and parental education, household financial resources, calendar time, and employment duration.

Table 7: Job-Quality-Standardized EU Hazard Ratios by Match Entry Origin

	E-origin matches Job-to-job entry	U-origin matches Entry from unemployment
<i>SP-to-TP</i> hazard ratio	1.1454** (0.0751)	0.9794 (0.0665)
Accepted matches	11,133	4,950
Full controls	Yes	Yes
Joint risk-week observations		1,650,920
EU transitions		2,960
Workers		4,923

Notes: The table reports *SP-to-TP* hazard ratios from a discrete-time complementary log-log model of weekly employment-to-unemployment transitions. Cell-specific adjusted hazards are aggregated using the pooled match-level distribution of wage terciles within each entry origin, with the same origin-specific weights applied to both family-structure groups. The specification includes demographic characteristics, own and parental education, parental and school-exit resources, calendar time, and employment duration. Delta-method standard errors of the reported hazard ratios are shown in parentheses and are clustered by worker. Significance levels are based on tests of $H_0 : HR = 1$. The p -value for equality of the two origin-specific hazard ratios is 0.053.

** $p < 0.05$.

I use estimated model to obtain adjusted weekly EU hazards for each family-structure $\times$ entry-path $\times$ wage-tercile cell. Within each entry path, I then aggregate the three tercile-specific predictions using a *common* distribution of observed job quality. The common weights are calculated from the pooled distribution of accepted matches across two family-structure groups, assigning one observation to each distinct worker-employer match. The same low, middle, high wage shares are therefore applied to workers from single-parent and two-parent households among jobs reached through job-to-job transitions and, separately, among jobs entered from unemployment. This standardization removes family-structure differences in the wage quality composition of accepted jobs without requiring the two entry paths to have the same quality distribution.

Table 7 reports the resulting *SP-to-TP* hazard ratios. Hazard ratios between two family-structure groups represents how much more likely *SP* workers are to experience that event. Among jobs reached through direct job-to-job transitions, the job-quality-standardized EU hazard is 14.5 percent higher for workers from single-parent households. The estimate is statistically significant at the 5 percent level ($p = 0.038$). Thus, equalizing the distribution of observed job quality does not eliminate the family-structure difference in subsequent employment stability among matches reached through employed search.

The corresponding pattern is not observed among jobs entered from unemployment. The standardized *SP*-to-*TP* hazard ratio is 0.979 and is not statistically distinguishable from 1 ($p = 0.759$). A direct test of equality between the two entry-path-specific hazard ratios yields a p -value of 0.053. This provides a suggestive evidence that the family-structure stability gap is larger among matches reached through job-to-job transitions.

The results therefore indicate different roles for observed job characteristics across entry paths. Among jobs entered from unemployment, the main family-structure difference lies in the quality distribution documented in Table 6. Once that quality distribution is equalized, there is no evidence of an additional EU hazard gap in jobs reached from unemployment. Among jobs reached through direct job-to-job transitions, by contrast, equalizing the distribution of observed job quality leaves a positive and statistically significant stability gap. This residual difference points to a dimension of match durability not summarized by the wage tercile of the accepted job.

The standardized comparison is an aggregate statement under a common job-quality distribution. It does not imply that the family-structure difference is uniform across all wage terciles. Appendix [X] reports the underlying within-tercile estimates and shows that the residual difference among jobs reached through job-to-job transitions is concentrated in the highest wage tercile, while none of the within-tercile comparisons among jobs entered from unemployment is statistically significant. I treat this pattern as supplementary heterogeneity rather than as an additional main empirical margin.

3.6 Summary of Empirical Findings

The empirical analysis establishes three main facts. First, workers from single parent households experience higher post-school unemployment even after conditioning on observed human capital, parental education, and household resources. The difference is driven primarily by a higher employment-to-unemployment hazard rather than by a lower unemployment-to-employment hazard.

Second, workers from single-parent households enter a less favorable distribution of accepted jobs. This difference is not confined to the initial school-to-work transition and is particularly pronounced among jobs obtained following unemployment. Thus, similar rates of reemployment

do not imply similar quality of reemployment.

Third, the family difference in subsequent employment stability varies with the path through which a job is entered. After standardizing for observed job quality distribution, the *SP-to-TP* EU hazard ratio is greater than 1 among jobs reached through direct job-to-job transitions, whereas the hazard ratio is close to one among jobs entered from unemployment.

These facts motivate a distinction between the *quality distribution* of job offer accessible from unemployment and the *stability distribution* of job offer accessible during employed search. The quantitative model formalizes this distinction by allowing family structure to shift the wage-quality distribution of opportunities accessible from unemployment and the stability composition of opportunities accessible while employed. These opportunity distributions, together with workers' acceptance decisions and job-ladder mobility, generate the patterns of accepted job and employment transition observed in the data.

4 Quantitative Job-Ladder Model

In this section, I develop a parsimonious job-ladder model to quantify how differences in job opportunity distribution can generate the family structure gap documented in Section 3. I build on the job-ladder framework of Baydur and Mukoyama (2020), extending their model of heterogeneous match characteristics to allow the distribution of quality and stability composition to differ by family structure. The model is not a model of family behavior or parental inputs. Family structure enters only as a predetermined group characteristic indexing a reduced-form difference in the distribution of job-offer. Conditional on the characteristics of a realized match, workers from the two family-background groups face the same separation technology.

Note that the model focuses on employment and unemployment and abstracts from labor-force nonparticipation. Workers may move from unemployment into employment, remain with their current employer, move directly to another employer, or separate into unemployment. Accordingly, the quantitative moments are constructed from worker-employer matches whose observed entry origins lie in employment or unemployment.⁵

⁵The full sample EU hazard estimates in Table 4 remain important as reduced-form facts, but they are not mechanically identical to model moments because the empirical sample also contains jobs entered from nonparticipation.

4.1 Environment

Time is discrete and measured in weeks. Workers belong to family structure group

$$g \in \{TP, SP\},$$

where TP denotes a two-parent household and SP denotes a single-parent household at age 16.

A worker is either unemployed, U , or employed, E in a match

$$m = (q, x, s).$$

The first match characteristic is observed wage quality,

$$q \in \{L, M, H\}.$$

It corresponds to the low, middle, and high terciles of the starting-wage distribution. Let w_q denote the normalized flow payoff associated with quality q .

The second component, x , captures nonwage match value. I assume

$$x \sim N(0, \sigma_x^2).$$

The mean-zero assumption centers x around the average match, so that x captures how the non-wage value of a particular match differs from that of an average match. Workers observe the realization of x when evaluating a job offer.

The third component,

$$s \in \{F, S\}$$

is latent match stability. Fragile matches separate into unemployment with weekly probability δ_F , while stable matches separate with probability δ_S , where

$$\delta_F > \delta_S.$$

Note that δ_F and δ_S are common across family structure groups. Family structure therefore does not directly alter the separation technology conditional on realized match type.

The timing at the offer stage follows Baydur and Mukoyama (2020). When a job offer arrives, the worker observes *all three* match attributes (q, x, s) before accepting or rejecting it.

An unemployed worker receives a job offer with probability λ_U and an employed worker receives with λ_E . I assume no voluntary move from employment to unemployment. For an employed worker in match $m = (q, x, s)$, events occur sequentially within the week. The current match first separates with probability δ_s . Conditional on survival, a job offer arrives with probability λ_E . If an offer arrives, the worker observes all of its characteristics (q', x', s') and accepts it only if the prospective match yields a higher continuation value than the current match.

4.2 Distributions of Match Characteristics

Let $m' = (q', x', s')$ be a newly arriving job offer. Note that the prime notation distinguishes the prospective match from a worker's current match $m = (q, x, s)$.

Conditional on an offer arriving with probability λ_o where current labor market state $o \in \{U, E\}$, m' follows joint probability distribution $G_o^g = P_{q,o}^g \otimes F_x \otimes \Pi_{s,o}^g$. Here, $P_{q,o}^g, F_x, \Pi_{s,o}^g$ represent probability measure governing wage quality q , nonwage match value x , and stability s , respectively.

The empirical restrictions are imposed through the following current labor state-specific distributions. Note that results from Table 6 guide us to let the wage-quality wedge of single-parent household appear only in U-origin jobs and results from Table 7 guide us to let the stability wedge of single-parent household appear only in E-origin jobs.

To be more specific, for offers received while unemployed,

$$dG_U^g(q', x', s') = p_U^g(q'; \alpha_U) f_x(x') \pi_U(s') dx'.$$

Family structure can therefore affect the wage quality composition of these offers through the parameter α_U , while the distributions of nonwage value and match stability are common across family types.

For offers received while employed,

$$dG_E^g(q', x', s' | q) = p_E(q' | q) f_x(x') \pi_E^g(s'; \gamma_E) dx'.$$

The transition kernel $p_E(q' | q)$ allows prospective quality to depend on current quality and captures persistence along the job ladder. It is common across family types. Family structure instead affects the stability composition of offers through γ_E . These distributions describe the composition of prospective matches conditional on an offer arriving. They are distinct from the offer-arrival probabilities λ_E and λ_U , which govern how frequently workers receive offers.

Section 5 specifies the functional form used for $p_U^g(q'; \alpha_U)$, $p_E(q' | q)$, and $\pi_E^g(s'; \gamma_E)$ and explains how the two family-specific margins are disciplined by the data.

4.3 Worker Values and Acceptance Decisions

Let V_U^g denote the value of unemployment for a worker from family structure group g , and let $V_E^g(m)$ denote the value of employment in match $m = (q, x, s)$. The flow payoff from unemployment is $\log b$, while the flow payoff from employment is

$$y(q, x) = \log w_q + x.$$

Unemployed Workers An unemployed worker receives a job offer with probability λ_U . Because the worker observes the prospective match before deciding, an offer m' is accepted if and only if

$$V_E^g(m') \geq V_U^g.$$

The unemployment value therefore satisfies

$$V_U^g = \log b + \beta \left[(1 - \lambda_U) V_U^g + \lambda_U \mathbb{E}_{m' \sim G_U^g} \max\{V_U^g, V_E^g(m')\} \right].$$

Define the probability of accepting an offer as

$$A_U^g = \Pr_{m' \sim G_U^g} [V_E^g(m') \geq V_U^g].$$

A UE transition thus requires both arrival of an offer and endogenous acceptance of that opportunity.

Employed Workers For an employed worker, events occur sequentially within the week. The current match first separates with probability δ_s . Conditional on survival, an offer arrives with probability λ_E . The worker observes the prospective match and moves to the new employer only if it yields a higher value than the current match. The employment value is

$$V_E^g(m) = \log w_q + x + \beta \left[\delta_s V_U^g + (1 - \delta_s) \left\{ (1 - \lambda_E) V_E^g(m) + \lambda_E \mathbb{E}_{m' \sim G_E^g(\cdot|q)} \max\{V_E^g(m), V_E^g(m')\} \right\} \right].$$

Define the probability of accepting the new offer as

$$A_E^g(m) = \Pr_{m' \sim G_E^g(\cdot|q)} [V_E^g(m') \geq V_E^g(m)].$$

Then the probability of a direct EE move from current match m is

$$(1 - \delta_s) \lambda_E A_E^g(m),$$

whereas the probability of remaining with the current employer is

$$(1 - \delta_s) [1 - \lambda_E A_E^g(m)].$$

4.4 Labor Market Dynamics

The model distinguishes the primitive events that generate observed labor market transitions. The acceptance probabilities derived in the previous subsection imply the following weekly transition

hazards:

$$h_{UE}^g = \lambda_U A_U^g, \quad h_{EU}(m) = \delta_s, \quad h_{EE}^g(m) = (1 - \delta_s) \lambda_E A_E^g(m).$$

For a worker in match m , an EU transition $h_{EU}(m)$ occurs with probability δ_s . An EE transition $h_{EE}^g(m)$ requires the current match to survive, an offer to arrive, and that offer to be accepted. A UE transition h_{UE}^g similarly requires both offer arrival and acceptance. This distinction allows observed transition rates to differ even when workers face common primitive arrival or separation parameters.

The distribution of workers across unemployment and heterogeneous matches evolves according to these transition probabilities. Given optimal acceptance rules, the model generates a group-specific stationary distribution over unemployment and employment states.

In the quantitative implementation, unemployment constitutes one state, while each discretized match $m = (q, x, s)$ constitutes a separate employment state. Let K^g denote the resulting Markov transition matrix for group g . Each row represents a worker's current state, and each column represents a possible state in the following week. The entries of K^g give the corresponding one-week transition probabilities, and each row sums to one.

Let μ_t^g denote the cross-sectional distribution of group g workers across these states at the beginning of week t , and let μ_0^g denote the initial distribution. The component $\mu^g(U)$ is the share of the group that is unemployed, while $\mu_t^g(m)$ is the share employed in match state m . Let μ^g denote the stationary distribution to which this cross-sectional distribution converges. Accordingly, $\mu^g(U)$ is the group's stationary unemployment share, while $\mu^g(m)$ is its stationary share in match state m .

The cross-sectional distribution evolves according to

$$\mu_{t+1}^g = \mu_t^g K^g,$$

which implies

$$\mu_t^g = \mu_0^g (K^g)^t.$$

Under the standard ergodicity condition, the stationary distribution can be characterized as the

long-run limit

$$\mu^g = \lim_{t \rightarrow \infty} \mu_0^g (K^g)^t.$$

Equivalently, it is the normalized fixed point satisfying

$$\mu^g = \mu^g K^g, \quad \mu^g(U) + \sum_m \mu^g(m) = 1.$$

Although the offer arriving probabilities and separation probabilities are common across family structure groups, their transition matrices and stationary distributions need not be. Differences in prospective match distributions affect which offers workers accept and which types of matches they accumulate along the job ladder.

This distinction is central to the models' explanation of the aggregate EU gap. The group-level EU hazard is an average of δ_F and δ_S , weighted by the stationary shares of employed workers in fragile and stable match. It can therefore differ across groups even though the separation probability associated with each stability type is identical.

The two family-specific wedges affect the composition of job offers through distinct margins. The quantitative implementation in Section 5 determines how these margins and remaining model parameters are parameterized and disciplined by the data.

5 Quantitative Implementation and Results

5.1 Parameterization and Calibration

I now specify the functional forms and numerical implementation used to take the model to the data.

Payoffs and wage quality I first express the empirical wage schedule in relative terms by normalizing the middle wage node to 1 by using the start-year wage terciles in the NLSY97 data.

The relative wage payoff nodes are

$$w_q = (w_L, w_M, w_H) = (0.60998, 1, 2.37765).$$

This normalization preserves the empirical low-to-middle and high-to-middle wage ratios, but it does not yet place employment and unemployment resources on a common scale. To establish that scale, I normalize unemployment resources, including the UI-equivalent component, to $b = 1$ and calibrate the unemployment-to-middle wage ratio as

$$\omega = \frac{b}{w_M} = 0.425.$$

This scale is based on the U.S. average of the Department of Labor's claimant-level UI Replacement Ratio 1.⁶ These two conditions imply

$$\tilde{w}_M = \frac{b}{\omega} = 2.35294.$$

I then preserve the empirical relative wage schedule when constructing the remaining employment-resource nodes:

$$\tilde{w}_q = \frac{w_q}{\omega} = (1.43524, 2.35294, 5.59448).$$

Distribution of wage-quality To parameterize ordered shifts in wage quality, define

$$Q_L = -1, \quad Q_M = 0, \quad Q_H = 1.$$

The index Q_q is not an additional observed job characteristic. It is a coding device that allows the model to parsimoniously represent a shift toward higher or lower wage quality states and distance between ordered states.

The baseline wage quality probabilities take the multinomial logit form:

$$p(q) = \frac{\exp(\eta_q)}{\sum_{k \in \{L, M, H\}} \exp(\eta_k)}, \quad \eta_M = 0.$$

The normalization $\eta_M = 0$ makes η_L and η_H the log odds of low and high quality relative to middle quality.

⁶The Department of Labor defines Replacement Ratio 1 as the weighted average of the weekly benefit amount divided by the claimant's normal hourly wage times 40 hours.

For offers received while unemployed, *TP* workers draw from the baseline distribution. The corresponding distribution for *SP* workers is

$$p_U^{SP}(q) = \frac{\exp(\eta_q + \alpha_U Q_q)}{\sum_{k \in \{L, M, H\}} \exp(\eta_k + \alpha_U Q_k)}.$$

A negative α_U shifts the wage-quality composition of opportunities accessible to *SP* workers toward lower-quality jobs. The parameter changes the composition of arriving offers, not their arrival frequency.

For offers received while employed, the wage-quality distribution also depends on the worker's current wage-quality state:

$$p_E(q' | q) = \frac{\exp(\{\eta_{q'} - \rho_E |Q_{q'} - Q_q|\})}{\sum_{k \in \{L, M, H\}} \exp(\{\eta_k - \rho_E |Q_k - Q_q|\})}$$

A positive ρ_E makes nearby wage quality states relatively more likely and therefore captures persistence along the job ladder. This kernel is common across family structure groups.

Distribution of nonwage match value Nonwage match value is assumed to follow normal distribution:

$$x \sim N(0, \sigma_x^2).$$

Its mean is normalized to zero, and the distribution is represented on a 51-point grid in the benchmark computation. I calibrate the dispersion σ_x using the fraction of direct EE wage quality changes that are upward,

$$Pr(q' > q | E, q' \neq q).$$

This moment identifies the strength of observed wage quality relative to unobserved nonwage value in workers' acceptance decision. When σ_x is small, accepted wage-quality changes are predominantly upward. As σ_x rises, a sufficiently favorable nonwage draw can compensate for a lower observed wage node.

The empirical conditional upward share is 0.59461. The calibration gives

$$\hat{\sigma}_x = 2.58791,$$

for which the model-implied share is 0.59486. I condition on $q' \neq q$ because the same-quality share of direct job-to-job transitions is used separately to discipline ρ_E , which is a quality persistence parameter.

Distribution of match stability I set the baseline stable-offer share to

$$\nu_0 = 0.5.$$

This baseline applies to offers received by *TP* workers while employed and to offers received by while unemployed by both family groups. For *SP* workers searching on the job, the stable share is

$$\nu^{SP} = \Pr(S \mid E, SP) = \Lambda [\text{logit}(\nu_0) + \gamma_E],$$

where

$$\Lambda(z) = \frac{1}{1 + \exp(-z)} \quad \text{and} \quad \text{logit}(p) = \log \frac{p}{1-p}.$$

Equivalently,

$$\log \frac{\nu^{SP}}{1 - \nu^{SP}} = \log \frac{\nu_0}{1 - \nu_0} + \gamma_E.$$

Thus γ_E is the *SP* shift in the log odds of accessing a stable rather than a fragile employed-search opportunity relative to the *TP* baseline. Because $\nu_0 = 0.5$, the baseline log odds are zero and the *SP* stable share simplifies to $\Lambda(\gamma_E)$. A negative γ_E therefore lowers the stable opportunity share for *SP* workers. Note that γ_E changes only the composition of accessible job offers; it does not alter the separation probabilities δ_S and δ_F associated with the realized stability types. The estimated values of the two family-structure wedges, α_U and γ_E , and their quantitative implications for the composition of accessible offers are presented in Section 5.2.

Regarding the relative separation risk of fragile and stable matches, I use the calibrated two-

type structure in Baydur and Mukoyama (2020) as the external benchmark. Although their empirical hazard analysis uses the NLSY79, the stability type-specific separation probabilities in their quantitative model are calibration objects rather than estimates of latent stability types from the NLSY79. Their average monthly separation probabilities for fragile and stable types are set to 0.021 and 0.011, respectively, with the aggregate EU flow disciplined by aggregate U.S. worker-flow evidence. I therefore do not interpret these numbers as cohort-specific estimates.

Because the our model operates at a weekly frequency, I convert the two monthly probabilities to weekly equivalents by preserving monthly survival:

$$1 - \delta_s^M = (1 - \delta_s^W)^{52/12}, \quad s \in \{F, S\}.$$

The resulting weekly-equivalent relative separation-risk benchmark is

$$r_\delta = \frac{1 - (1 - 0.021)^{12/52}}{1 - (1 - 0.011)^{12/52}} = 1.91654.$$

Only this relative separation risk spread is taken externally. The weekly stable-type separation probability δ_S is estimated jointly with the remaining structural parameters from the NLSY97, and the fragile-type probability is implied by $\delta_F = r_\delta \delta_S$.

I set the annual discount factor 0.96 and convert it to the weekly frequency. The benchmark unemployment payoff is set to $b = 1$ on the normalized payoff scale; Section ?? varies the SP unemployment payoff directly.

Table 8 summarizes the quantities held fixed outside the joint minimum-distance estimation. The relative wage nodes and the dispersion of nonwage match value are disciplined by the NLSY97, whereas the remaining quantities are external inputs, benchmark assumptions, normalizations, or quantities implied by them. In particular, the unemployment resource level is normalized to $b = 1$, and the empirical relative wage nodes are rescaled into the employment-resource nodes used in the log utility specification.

Table 8: Calibrated Inputs, Benchmark Assumptions, and Normalizations

Quantity	Value	Source or quantitative role
<i>Panel A. Data-based inputs held fixed outside the estimator</i>		
σ_x	2.58731	NLSY97 directional wage-quality changes among direct EE moves
(w_L, w_M, w_H)	(0.60998, 1, 2.37765)	NLSY97 starting-wage distribution; middle node normalized to 1
<i>Panel B. Benchmark and external inputs</i>		
$\omega = b/w_M$	0.425	U.S. Department of Labor unemployment-insurance data; benchmark ratio of unemployment resources to middle wage
$r_\delta = \delta_F/\delta_S$	1.91654	Baydur and Mukoyama (2020) benchmark, converted to weekly frequency
ν_0	0.50	Equal stable and fragile baseline offer shares
β	$0.96^{1/52} = 0.999215$	Weekly equivalent of an annual discount factor of 0.96
<i>Panel C. Normalizations and implied resource scale</i>		
b	1	Unemployment-resource normalization
η_M	0	Reference category for wage-quality log weights
$\mathbb{E}[x]$	0	Location normalization for nonwage match value
$(\tilde{w}_L, \tilde{w}_M, \tilde{w}_H)$	(1.43524, 2.35294, 5.59448)	Employment-resource nodes implied by $\tilde{w}_q = w_q/\omega$

Notes: Data-based inputs in Panel A are constructed from the NLSY97 but are held fixed outside the joint minimum-distance estimation. The benchmark utility specification is $u(c) = \log c$, so the normalization $b = 1$ implies $u(b) = 0$; the resource level itself is 1, not zero. The employment-resource nodes are the quantities entering $u(\tilde{w}_q) + x$.

5.2 Estimation and Targeted Fit

The eight free parameters are estimated by minimum distance: $\theta = (\lambda_U, \lambda_E, \eta_L, \eta_H, \rho_E, \delta_S, \alpha_U, \gamma_E)'$.

Let $\hat{m}_j$ denote the empirical target j , $m_j(\theta)$ its model counterpart, and s_j the empirical scale used to standardized the discrepancy. The minimum-distance objective is

$$\hat{\theta} = \arg \min_{\theta \in \Theta} Q(\theta),$$

where

$$Q(\theta) = \sum_{j=1}^J d_j \left[\frac{m_j(\theta) - \hat{m}_j}{s_j} \right]^2,$$

where d_j denotes the block weight.

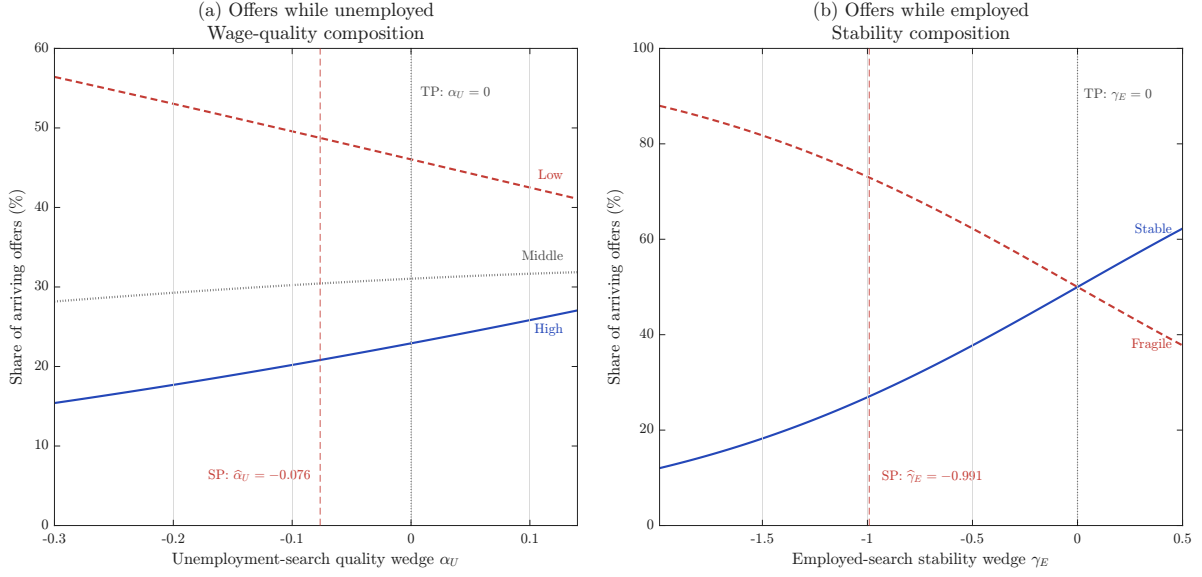
Table 9: Estimated Parameters and Their Empirical Discipline

Parameter	Estimate	Interpretation and empirical discipline
λ_U	0.030871	Common weekly job-offer arrival probability while unemployed; disciplined by the pooled UE transition probability through endogenous offer acceptance
λ_E	0.042636	Common weekly job-offer arrival probability while employed; disciplined by the pooled direct EE transition probability through current-match survival and endogenous offer acceptance
η_L	0.394387	Log weight on low-quality offers relative to the middle-quality node; disciplined primarily by the low-quality share among <i>TP</i> jobs reached through direct job-to-job transitions
η_H	−0.303577	Log weight on high-quality offers relative to the middle-quality node; disciplined primarily by the high-quality share among <i>TP</i> jobs reached through direct job-to-job transitions
ρ_E	0.715014	Persistence of wage quality in offers received while employed; disciplined by the same-quality share of direct job-to-job transitions
δ_S	0.00134590	Weekly separation probability of a stable match; disciplined by the <i>TP</i> EU transition probability through the stationary composition of stable and fragile matches
α_U	−0.076425	<i>SP</i> shift in the quality composition of offers received while unemployed; disciplined by <i>SP</i> – <i>TP</i> differences in the quality composition of jobs entered from unemployment
γ_E	−0.991114	<i>SP</i> shift in the log odds of receiving a stable rather than fragile offer while employed; disciplined by the quality-standardized <i>SP</i> / <i>TP</i> EU hazard ratio for jobs reached through direct job-to-job transitions

Notes: The parameters λ_U and λ_E are primitive job-offer arrival probabilities, not UE and EE transition probabilities. A UE transition also requires offer acceptance, while a direct EE transition requires current-match survival and offer acceptance. Similarly, δ_S is the separation probability conditional on a stable match rather than the aggregate *TP* EU probability. The pooled UE and direct EE targets combine *TP* and *SP* workers using their empirical shares in the relevant unemployment and employment risk sets. All probabilities are weekly.

The eleven core targets are chosen to discipline distinct parts of the model: the weekly UE and direct EE probabilities pooled across family-structure groups; the *TP* weekly EU probability; the low and high quality shares among *TP* jobs reached through direct job-to-job transitions; the quality-standardized *SP*/*TP* EU hazard ratio; the low and high quality shares among *SP* and *TP* jobs entered from unemployment; and the same-quality share of direct job-to-job transitions. Here, pooled means that *TP* and *SP* workers are combined for the corresponding transition target. The model counterpart averages the group-specific transition probabilities using the empirical shares of *TP* and *SP* workers in the relevant unemployment or employment risk set.

Figure 2: Family Structure and Accessible Job-Offer Distributions



Notes: Panel (a) shows how the U-origin quality wedge α_U shifts the conditional distribution of arriving offers across low-, middle-, and high-quality categories. The *TP* benchmark is normalized to $\alpha_U = 0$, while the estimated *SP* wedge is $\hat{\alpha}_U = -0.076$. Panel (b) shows how the E-origin stability wedge γ_E changes the conditional shares of stable and fragile offers. The *TP* benchmark satisfies $\lambda_{E,S}^{TP} = \nu_0 = 0.5$, corresponding to $\gamma_E = 0$, while the estimated *SP* wedge is $\hat{\gamma}_E = -0.991$. The figure describes accessible-offer distributions conditional on an offer arriving, rather than accepted-job distributions.

Table 9 reports the eight parameters jointly estimated by minimum distance. The first six parameters govern common offer arrival, wage-quality, persistence, and separation margins. The final two, α_U and γ_E , are the family-specific offer-access wedges. Figure 2 and 3 translate these into estimates into their implications for the composition of accessible offers. The estimated family-specific margins are

$$\hat{\alpha}_U = -0.076, \quad \hat{\gamma}_E = -0.991.$$

The first shifts *SP* unemployment-search offers towards lower wage quality. Under the calibration $\nu_0 = 0.5$, the second implies

$$\hat{\nu}^{SP} = \Pr(S \mid E, SP) = \Lambda(\hat{\gamma}_E) \simeq 0.270,$$

compared with 0.5 for *TP* employed search. This stable share is a latent model-implied object conditional on the benchmark stability structure, not a directly observed probability.

Table 10 compares the empirical targets with their model counterparts evaluated at $\hat{\theta}$. The

Figure 3: Model-Implied Accessible Job-Offer Distributions by Family Structure



Offer shares while unemployed (%)

Worker type	Low	Middle	High
TP	46.05	31.04	22.91
SP	48.74	30.44	20.82

Offer shares while employed (%)

Worker type	Fragile	Stable
TP	50.00	50.00
SP	72.93	27.07

Notes: Panel (a) reports the model-implied distribution of U-origin offers across wage-quality categories. The TP distribution is evaluated at the normalization $\alpha_U = 0$, while the SP distribution is evaluated at $\hat{\alpha}_U = -0.076$. Panel (b) reports the corresponding distribution of E-origin offers across fragile and stable opportunities. The TP distribution is evaluated at $\gamma_E = 0$, while the SP distribution is evaluated at $\hat{\gamma}_E = -0.991$. All shares are conditional on an offer arriving and therefore describe accessible-offer distributions rather than accepted-job distributions.

Table 10: Targeted Model Fit

Target moment	Data	Model
Pooled weekly UE transition probability	0.023421	0.023419
Pooled weekly direct EE transition probability	0.003677	0.003676
<i>TP</i> weekly EU transition probability	0.001742	0.001741
<i>TP</i> job-to-job entries: low-quality share	0.254739	0.249805
<i>TP</i> job-to-job entries: high-quality share	0.428020	0.422688
Job-to-job entries: quality-standardized <i>SP/TP</i> EU hazard ratio	1.164115	1.163896
Entry from unemployment, <i>SP</i> : low-quality share	0.379531	0.396647
Entry from unemployment, <i>TP</i> : low-quality share	0.345646	0.365282
Entry from unemployment, <i>SP</i> : high-quality share	0.258193	0.267754
Entry from unemployment, <i>TP</i> : high-quality share	0.285410	0.293167
Direct job-to-job transitions: same-quality share	0.552770	0.553155

Notes: The pooled UE and direct EE moments combine *TP* and *SP* workers using their empirical shares in the relevant risk set. The quality-standardized family difference is displayed as a hazard ratio for consistency with the empirical tables, but its estimation target is computed on the log-hazard-ratio scale. Transition probabilities are weekly, and probabilities and shares are reported as fractions. The minimum-distance criterion at the reported estimate is $Q(\hat{\theta}) = 1.530$; it is a measure of model fit rather than a formal overidentification statistic.

minimum-distance objective evaluated at the estimated parameter vector is $Q(\hat{\theta}) = 1.530$. All eleven targeted model-data discrepancies are smaller than on empirical error scale in absolute value. The model closely reproduces pooled UE and EE transition probabilities, the *TP* EU probability, the persistence of wage quality across direct EE moves, and the quantity-standardized E-origin family difference in employment exits. Note that the reported job-quality shares describe the composition of *realized* jobs rather than the composition of all offers received.

5.3 Untargeted Moments Validation

I next examine whether the estimated model reproduces aggregate labor market outcomes beyond those directly fitted by the minimum-distance criterion. Panel A of Table 11 compares stationary unemployment rates. The model generates rates of 6.856 percent for *TP* workers and 7.887 percent for *SP* workers, implying $SP - TP$ gap of 1.031 percentage points. The corresponding empirical rates are 6.31 and 6.91 percent, with a gap of 0.61 percentage point. Thus, the model reproduces the direction and broad magnitude of the unemployment difference, although it overstates the gap by approximately 0.42 percentage point.

Table 11: Validation of Aggregate Labor-Market Outcomes

Outcome	Data benchmark	Model
<i>Panel A. Stationary unemployment rate</i>		
TP workers	6.31%	6.86%
SP workers	6.91%	7.89%
SP – TP gap	0.61 pp	1.03 pp
<i>Panel B. Weekly employment-to-unemployment transitions</i>		
TP workers	0.1742%	0.1741%
SP workers	0.1963%	0.1974%
SP/TP EU hazard ratio	1.105	1.134

Notes: TP and SP denote workers from two-parent and single-parent family households, respectively. In Panel A, the data benchmarks are standardized predicted unemployment probabilities from the fully controlled empirical specification, averaged over a common covariate distribution. The model column reports the unemployment mass in each group’s stationary distribution. Stationary unemployment is not targeted in estimation. In Panel B, probability levels are expressed in percent per week. The empirical hazard ratio is the exponentiated SP coefficient from the fully controlled complementary log-log specification reported in Table 4. The model hazard ratio is computed from the aggregate weekly transition probabilities as $[-\log(1 - h_{EU}^{SP})]/[-\log(1 - h_{EU}^{TP})]$. Because the empirical hazard ratio is a regression-based conditional object, it need not equal the ratio of the displayed standardized probability levels. The TP weekly EU probability is an estimation target; the aggregate SP probability and aggregate SP/TP hazard ratio are untargeted validation outcomes. Differences may not equal differences between displayed rounded values.

Panel B provides a further validation using weekly EU transitions. The TP weekly EU probability is included among the estimation targets and therefore serve as the baseline anchor. The aggregate SP probability and the aggregate SP/TP hazard ratio, however, are not targeted; the family-specific EU target is instead the job-quality-standardized SP/TP hazard ratio among jobs reached through direct job-to-job transitions. The model implies a weekly EU probability of 0.1974 percent for SP workers, compared with 0.1963 percent in the data. Applying the complementary log-log transformation used in the empirical analysis, the model probabilities imply an aggregate SP/TP hazard ratio of 1.134, compared with 1.105 in Table 4. Thus, the model implies that SP workers face a 13.4 percent higher aggregate EU hazard, compared with 10.5 percent in the data. Although the model somewhat overstates the relative difference, it reproduces its direction and approximate magnitude. This comparison provides an untargeted check that the estimated offer-access wedges and the resulting stationary match composition translate into the observed difference in aggregate employment instability.

5.4 Mechanism Decomposition

The model generates stationary unemployment rates of 6.86 percent for *TP* workers and 7.89 percent for *SP* workers, a gap of 1.03 percentage points. The corresponding weekly EU probabilities are 0.1741% and 0.1985%.

I next conduct a mechanism decomposition to separate the roles of the two family-specific offer-access margins. For each outcome, I recompute the stationary equilibrium after setting $\alpha_U = 0$, setting $\gamma_E = 0$, and setting both wedges to zero. Because the model is nonlinear, the contribution assigned to one wedge can depend on whether the other wedge has already been removed. I therefore average each wedge's marginal contribution over the two possible equalization orders, following Shapley decomposition (Shapley et al., 1953; Shorrocks et al., 2013). Appendix D provides the formal construction. These decomposition exercises diagnose the sources of model-generated gaps rather than the effects of feasible policies: setting a wedge to zero mechanically removes the primitive family difference represented by that wedge.

Panel A of Table 12 shows that the employed-search stability wedge accounts for nearly all of the model-generated differences in aggregate employment instability. The benchmark model generates an aggregate weekly *SP-TP* EU probability gap of 0.0233 percentage point, corresponding to an *SP/TP* hazard ratio of 1.134. The employed-search stability wedge accounts for 100.04 percent of this gap. The unemployment-search quality wedge contributes -0.04 percent and therefore provides only a negligible offset. The same separation holds for the job-quality-standardized family difference among jobs reached through job-to-job transitions. The benchmark hazard ratio is 1.1639; on the log-hazard scale used for the additive decomposition, the stability wedge accounts for 100.04 percent of the difference, while the quality wedge contributes -0.04 percent.

The stability margin also accounts for most of the stationary unemployment difference. The benchmark model produces an *SP-TP* stationary unemployment gap of 1.0307 percentage points. The stability wedge contributes 104.43 percent of this gap, whereas the quality wedge contributes -4.43 percent. The contribution above 100 percent means that the stability wedge by itself would generate a slightly larger unemployment gap and that the quality wedge modestly offsets it. Thus, the model attributes both higher aggregate employment-exit risk and higher stationary unemploy-

Table 12: Mechanism Decomposition of Family-Structure Gaps

Outcome	$SP-TP$ gap	Contribution (% of gap)	
		Quality wedge (α_U)	Stability wedge (γ_E)
<i>Panel A. Aggregate labor-market outcomes</i>			
Weekly EU transition probability	0.0233 pp	−0.04	100.04
Stationary unemployment rate	1.0307 pp	−4.43	104.43
<i>Panel B. Stationary employment composition</i>			
Low-quality employment share	2.1697 pp	46.65	53.35
High-quality employment share	−1.9186 pp	53.63	46.37

Notes: Gaps are SP minus TP and are measured in percentage points. The last two columns report order-averaged Shapley contributions obtained by equalizing α_U and γ_E separately and jointly. The weekly EU probability gap corresponds to an SP/TP hazard ratio of 1.134. None of the reported gaps is directly targeted. Negative contributions indicate offsets; contributions above 100 percent arise when the other wedge moves the outcome in the opposite direction. These mechanism decompositions provide model accounting rather than policy effects.

ment primarily to reduced access to stable opportunities while employed.

Before turning to the long-run employment distribution, it is useful to distinguish the composition of entry flows from that of the employment stock. The job-quality composition immediately following unemployment provides a specification check rather than an independent mechanism result. By construction, α_U is the only wedge that directly changes the quality distribution of offers received while unemployed, whereas γ_E changes the stability composition of offers received while employed. Accordingly, α_U accounts for essentially all of the model-generated low and high quality gaps among jobs entered from unemployment. Any contribution of γ_E to these entry quality gaps is indirect, arising because future on-the-job opportunities enter employment values and therefore affect acceptance decisions at reemployment.

Panel B presents the propagation result by decomposing the stationary job-quality composition of employment. Unlike the job quality composition immediately following unemployment, the stationary employment distribution is not directly targeted and incorporates the cumulative effects of acceptance, separation, reemployment, and direct job-to-job mobility. The model generates an $SP-TP$ gap of 2.17 percentage points in the low quality employment share and −1.92 points in the high quality share. The unemployment-search quality wedge accounts for 46.65 percent of the low quality gap and 53.63 percent of the high quality gap, while the employed-search stability wedge accounts for the remaining 53.35 and 46.37 percent, respectively.

Table 13: Summary of Policy Counterfactuals

Counterfactual	First reported parity crossing		
	Stationary unemployment	Aggregate EU	Standardized Job-to-Job EU
Unemployment-resource support			
$\tilde{b}^{SP} = (1 + \tau)b$	Not reached	Not reached	Not reached
Generic employed-search access	$m_E = 1.24$	$m_E = 20.00$	
$\tilde{\lambda}_E^{SP} = m_E \lambda_E$	(24%)	(1,900%)	Not reached
Targeted stable access			
$\tilde{\nu}^{SP} = m_S \nu^{SP}$	$m_S = 1.50$	$m_S = 2.59$	$m_S = 2.66$

Notes: Entries report the first parity crossing on each discrete counterfactual grid. Percentages in parentheses are changes in total employed-search access relative to its *SP* baseline. For unemployment-resource support, no parity outcome is reached over $\tau \in [0, 4]$. “Standardized J2J EU” is the job-quality-standardized *SP/TP* weekly EU hazard ratio among jobs reached through direct job-to-job transitions. The crossing at $m_E = 20$ is a coarse-grid result rather than an interpolated threshold.

5.5 Policy Counterfactuals

The preceding subsection used counterfactual wedge equalization as a model-accounting exercise. By mechanically setting α_U or γ_E to zero, the decomposition identified how much of each model-generated family-structure gap is attributable to the unemployment-search quality margin and the employed-search stability margin. These exercises were deliberately diagnostic rather than policy-based: the estimated wedges summarize differences in opportunity access associated with family structure, but they are not themselves parameters that a policymaker can directly set to zero. gaps.

This subsection instead how much employment instability can be reduced by changing margins that policy could feasibly affect while leaving the estimated family-structure wedges in place. I consider counterfactual changes in three such margins: resources available while unemployed, generic access to job offers while employed, and access specifically to stable opportunities. Table 13 summarizes the resulting parity thresholds, while Table reports selected points along each counterfactual response path.

Note that, to keep the notation consistent, I use superscripts for family structure, $g \in \{TP, SP\}$, and subscripts for labor-market origin and, when needed, stability type. Thus, λ_E^g denotes the total employed-search job-offer arrival probability for group g , while ν^g denotes its stable component.

5.5.1 Unemployment Resource Support

The first exercise increases the *SP* unemployment resources available to *SP* workers,

$$\tilde{b}^{SP} = (1 + \tau)b, \quad \tau > 0, \quad b^{TP} = b,$$

while holding job offer arrival rates and offer distributions fixed.

Higher unemployment resources raise the value of remaining unemployed and make *SP* workers more selective about offers received while unemployed. This selection channel improves the job quality composition of accepted offers. As Panel A of Table 14 shows, increasing τ from zero to four reduces the low quality share among jobs entered from unemployment from 39.66 to 32.82 percent and raises the high quality share from 26.78 to 34.63 percent.

This improvement in accepted job quality does not translate into greater employment continuity. The stationary unemployment gap instead rises from 1.0307 to 4.5991 percentage points because the higher value of unemployment lowers acceptance and lengthens unemployment spells. Both employment-exit hazard ratios also remain above 1 throughout the reported grid. The counterfactual therefore reaches neither stationary unemployment parity, aggregate EU parity, nor job-quality standardized EU parity among jobs reached through job-to-job transitions.

5.5.2 Generic On-the-Job Search Support

The second exercise increases the *frequency* with which *SP* workers receive job offers while employed:

$$\tilde{\lambda}_E^{SP} = m_E \lambda_E, \quad m_E > 1,$$

holding the wage quality and stability composition of those offers fixed. Equivalently, the arrival rates of stable and fragile offers increase by the same proportion. The counterfactual therefore expands the quantity of on-the-job opportunities without directly changing their composition.

The stationary unemployment gap first closes on the policy grid at $m_E = 1.24$, corresponding to a 24 percent increase in total employed-search access. Because stable and fragile arrivals increase proportionally, the stable share among offers remains at its baseline *SP* value of 0.2707. The

Table 14: Selected Counterfactual Response Paths

<i>Panel A. Unemployment-resource support</i>					
τ	Low quality share (%)	High quality share (%)	Stationary U gap (pp)	Aggregate EU HR	Standardized J2J EU HR
0	39.66	26.78	1.0307	1.1341	1.1639
0.50	37.58	28.53	1.6886	1.1378	1.1712
1.00	36.06	30.16	2.3024	1.1372	1.1724
2.00	34.64	32.40	3.0899	1.1348	1.1715
4.00	32.82	34.63	4.5991	1.1357	1.1767
<i>Panel B. Generic employed-search access</i>					
m_E	Total access change (%)	Stable-offer share	Stationary U gap (pp)	Aggregate EU HR	Standardized J2J EU HR
1.00	0	0.2707	1.0307	1.1341	1.1639
1.24	24	0.2707	-0.0213	1.1408	1.1687
2.00	100	0.2707	-0.8405	1.1350	1.1576
5.00	400	0.2707	-0.8989	1.1233	1.1406
10.00	900	0.2707	-1.1601	1.0710	1.0880
20.00	1,900	0.2707	-1.5395	0.9956	1.0426
<i>Panel C. Targeted access to stable opportunities</i>					
m_S	Total access change (%)	Stable-offer share	Stationary U gap (pp)	Aggregate EU HR	Standardized J2J EU HR
1.00	0	0.2707	1.0307	1.1341	1.1639
1.50	13.53	0.3576	-0.0316	1.0809	1.0974
2.00	27.07	0.4261	-0.7700	1.0365	1.0452
2.59	43.04	0.4901	-1.2341	0.9994	1.0040
2.66	44.93	0.4968	-1.3090	0.9952	0.9996

Notes: Entry shares in Panel A refer to jobs entered from unemployment. The stationary- U gap is the $SP-TP$ difference in percentage points. The EU columns report SP/TP weekly hazard ratios. “Standardized J2J” denotes the job-quality-standardized EU hazard ratio among jobs reached through direct job-to-job transitions. Parity is zero for the unemployment gap and 1 for the hazard ratios. Bold entries mark the first reported grid crossing for each outcome. Access changes in Panels B and C are measured relative to the baseline employed-search arrival rate. The stable-offer share is fixed in Panel B and changes endogenously in Panel C.

primitive stability disadvantage is therefore unchanged.

Panel B of Table 14 shows that the employment-exit effects are both indirect and nonmonotonic at relatively small multipliers. At $m_E = 1.24$, the stationary unemployment gap has closed, but the aggregate and job-quality standardized EU hazard ratios are 1.1408 and 1.1687, respectively, slightly above their baseline values. Both hazard ratios eventually declines as employed-search access becomes much larger, but the adjustment is slow because the stability composition of arriving offers remains fixed.

The job-quality standardized employment-exit gap among jobs reached through direct job-to-

job transitions does not close over the reported grid. Aggregate EU parity is first reached only at $m_E = 20$, when total employed-search access is 20 times its baseline level, an increase of 1,900 percent. Even at this point, the job-quality standardized direct job-to-job EU hazard ratio remains 1.0426. The aggregate EU threshold is the first crossing on a coarse upper grid and should not be interpreted as a precise policy requirement. Instead, it demonstrates that expanding generic offer access is an extremely indirect way to eliminate the employment-exit gap when the stability composition of offers remains fixed.

5.5.3 Targeted Access to Stable Opportunities

The third exercise targets access to stable job offers for SP workers. Recall that $v^{SP} = 0.270692$ is the baseline stable share among offers received by SP workers while employed. Because λ_E is the total employed-search offer arrival rate, stable and fragile offers arrive at baseline rates $v^{SP}\lambda_E$ and $(1 - v^{SP})\lambda_E$, respectively.

For $m_S > 1$, the counterfactual scales only the arrival rate of stable offers:

$$v^{SP}\lambda_E \longrightarrow \tilde{v}^{SP}\lambda_E = m_S v^{SP}\lambda_E.$$

Thus, $m_S = 1$ corresponds to the baseline, while $m_S > 1$ adds stable opportunities without removing any fragile opportunities. The resulting total employed-search arrival rate and stable-offer share are, respectively,

$$\tilde{\lambda}_E^{SP} = \lambda_E \left[1 + (m_S - 1)v^{SP} \right] \quad \text{and} \quad \tilde{v}^{SP} = \frac{m_S v^{SP}}{m_S v^{SP} + 1 - v^{SP}}.$$

Unlike the generic employed-search counterfactual, which raises stable and fragile arrival rate proportionally, this exercise changes *both* the quantity and stability composition of accessible opportunities. The estimated family-structure wedge is not set to zero; instead, additional stable opportunities are added on top of the baseline opportunity distribution implied by the wedge.

The stationary unemployment gap first crosses zero on the reported grid at $m_S = 1.50$. A 50 percent increase in the stable-offer arrival rate raises total employed-search access by only 13.53

percent because stable offers initially account for 27.07 percent of *SP* opportunities. The stable share among arriving offers consequently rises from 0.2707 to 0.3576.

The employment-exit hazards respond more directly than under generic employed-search support. $m_S = 2.00$, total employed-search access has increased by 27.07 percent, while the aggregate and job-quality standardized direct job-to-job EU hazard ratios have already fallen from their baseline values of 1.1341 and 1.1639 to 1.0365 and 1.0452, respectively.

Aggregate EU parity is first crossed on the reported grid at $m_S = 2.59$. At this point, stable-offer arrivals are 159 percent above baseline, but total employed-search access is only 43.04 percent higher because fragile-offer arrivals remain unchanged. The stable-offer share rises to 0.4901, and the aggregate EU hazard ratio falls to 0.9994. The job-quality standardized direct job-to-job EU hazard ratio remains slightly above parity at 1.0040.

The standardized direct job-to-job EU hazard ratio first crosses parity at $m_S = 2.66$. This multiplier represents a 166 percent increase in stable-offer arrivals, a 44.93 percent increase in total employed-search access, and an implied stable-offer share of 0.4968. The corresponding standardized hazard ratio is 0.9996. The contrast with generic employed-search support therefore shows that the stability composition of accessible opportunities, rather than offer quantity alone, is central to reducing the model-generated employment-exit gap.

Taken together, the three policy counterfactual exercises distinguish three margins: resources available while unemployed, the quantity of offers available while employed, and the stability composition of those offers. Additional unemployment resources improve the job-quality composition of accepted offers but lengthen unemployment and do not repair the stability margin. Generic employed-search support can close the stationary unemployment gap with a moderate increase in total access, but it leaves the job-quality standardized employment-exit difference unchanged over the reported grid and closes the aggregate EU gap only at an extremely large arrival rate multiplier. By contrast, targeted stable access acts directly on the mechanism generating employment instability and closes both aggregate and standardized EU gaps with substantially smaller changes in total employed-search access. These numerical magnitudes are model-generated benchmarks and should not be interpreted as welfare rankings or causal effects of particular programs.

6 Conclusion

This paper studies how family structure during adolescence is associated with employment instability in young adulthood. Using weekly labor market histories from the NLSY97, I show that the higher post-school unemployment experienced by workers from single-parent households is driven primarily by more frequent employment-to-unemployment transition rather than by slower unemployment-to-employment transition rates. The evidence points to two distinct margins along the job ladder: these workers enter a less favorable distribution of jobs following unemployment, and they experience lower subsequent match durability among jobs reached through direct job-to-job transitions even after accounting for observed job quality. A quantitative job-ladder model represents these patterns as differences in the quality composition of opportunities accessible from unemployment and the stability composition of opportunities accessible while employed. The model attributes most of the aggregate employment-exit and stationary unemployment gaps to the stability margin, while the quality margin directly shapes the jobs reached after unemployment and propagates through subsequent labor market transitions. Within the model, expanding access specifically to stable opportunities reduces employment instability more directly than either increasing unemployment resources or expanding the overall quantity of on-the-job offers. The composition of accessible opportunities, not only the rate at which they arrive, is therefore an important margin of early-career employment inequality.

The paper makes three main contributions. First, it brings a flow-based perspective to the relationship between family structure and young-adult labor market outcomes. Rather than treating unemployment as a static outcome, the analysis identifies the transitions through which family-structure differences accumulate. Second, it shows that family-structure differences extend *beyond* educational attainment and the probability of finding a job. The composition and subsequent durability of accepted matches provide additional dimensions of early-career inequality. In particular, distinguishing jobs entered from unemployment from those reached through direct job-to-job transitions reveals different family-structure patterns that would be obscured in a pooled sample of accepted jobs. Third, the quantitative framework separates two proximate margins of opportunity access and traces how they interact through acceptance, separation, reemployment,

and job-to-job mobility. This provides an accounting framework for connecting entry conditions to longer-run differences in employment continuity.

The principal limitation of the analysis is that it locates the labor market margins at which family-structure differences emerge but does not *identify* their ultimate causes. The quality and stability wedges are reduced-form accounting objects that summarize differences in opportunity access; they are not themselves deep economic mechanisms. The paper therefore shows where along the job ladder the family-structure gap appears, but not why workers from different family structure face different quality and stability distributions. These difference could reflect unobserved cognitive or noncognitive human capital, family and social networks, geographic or transportation constraints, parental time and financial support, employer screening, or other frictions correlated with adolescent family structure. The supplementary analysis using observed noncognitive characteristics, parental proximity, and grandparent availability provide descriptive evidence on several possible explanations, but the available measure cannot distinguish among them or establish causal mediation. Accordingly, the results should not be interpreted as the causal effect of growing up in a single-parent household.

The model is also deliberately parsimonious. Its latent stability margin captures the overall job-quality standardized employment-exit difference among jobs reached through job-to-job transitions, but it does not account for the empirical concentration of this difference in the highest observed job-quality tercile. Explaining that heterogeneity may require a richer interaction between observed job quality and latent match stability. In addition, the NLSY97 observes accepted jobs rather than rejected offers. The estimated wedges therefore combine potentially distinct sources of limited access and should not be interpreted as literal firm-side offer distributions. Finally, the counterfactuals hold the estimated environment fixed and do not incorporate firm responses, general-equilibrium effects, parameter uncertainty, or welfare comparisons. They should be read as model-generated benchmarks for comparing margins rather than as estimates of the causal effects of particular programs.

These limitations point to a natural direction for future research: identifying the underlying factors that generate unequal access to high-quality and durable matches. Data connecting employment histories to family transfers, social networks, geographic constraints, employer charac-

teristics, and rejected as well as accepted opportunities would help distinguish among the possible channels. More broadly, the results suggest that understanding early-career inequality requires looking beyond whether and how quickly young workers find employment. The types of jobs they can effectively access, and whether those jobs provide durable employment, may be equally important for the accumulation of stable labor market experience.

Appendix A Data Construction and Additional Measures

Appendix B Robustness and Alternative Explanations

B.1 Observed Noncognitive Characteristics

The main results from Table 3 leaves a remaining association between family structure and unemployment after conditioning on observed education and family financial resources. One possible concern is that this association may partly reflect persistent worker characteristics that are not included in the main specification. I examine this possibility using the noncognitive measures available in the NLSY97.

Measure construction I use two sets of noncognitive measures from the NLSY97. The first is based on eight Goldberg personality items collected in Round 6. These items describe respondents along dimensions including organization, conscientiousness, dependability, thoroughness, agreeableness, cooperativeness, flexibility, and trustfulness. I recode these items so that higher values consistently indicate more of the relevant trait and then I combine them into measures of conscientiousness and agreeableness. Each measure is converted into a z-score by subtracting its sample mean and dividing by its sample standard deviation.

The second measure is an earlier respondent-reported behavioral and emotional problems scale from Round 1. Higher values of this measure indicate more behavioral or emotional problems. This measure is distinct from the Goldberg personality items and was collected substantially earlier in the respondents' lives.

The measures used here is different from those in Cunha et al. (2006), whose NLSY79 analysis identifies a noncognitive factor using the Rosenberg Self-Esteem and Rotter Locus-of-Control scales. My objective is not to replicate their latent-factor measure, but to use the earlier noncognitive characteristics available in the NLSY97 as proxies for dimensions of persistent worker heterogeneity that may be relevant for labor market outcomes.

The Goldberg items were collected in 2002–2003 and are available only for the younger among NLSY97 birth cohorts. To ensure that the personality measures *precede* the labor market outcomes

used in this exercise, I restrict the outcome window to 2006 and later. This restriction, together with the limited cohort coverage of the Goldberg module, substantially reduces the number of respondents from 7347 in the main analysis sample to 4029 in the Goldberg sample.

Empirical design I re-estimate the post-school labor force unemployment specification used in the main analysis. The dependent variable equals 1 in an unemployed week and 0 in an employed week. All specifications include the full set of controls used in the most controlled unemployment specification in the main text: demographic, own education, parental education, calendar quarter fixed effects, and family financial resources.

Because the Goldberg measures are observed only for a restricted subset of respondents, I separate changes induced by sample restriction from changes induced by adding the noncognitive controls. I first compare the average marginal effect (AME) of single-parent household in the main analysis sample with the corresponding estimate in the Goldberg sample. I directly test whether the estimated family-structure association differs from the main and Goldberg sample. Because the Goldberg sample is a subset of the main sample, the comparison accounts for the fact that many of the same respondents contribute to both estimates. I implement this comparison by estimating the two specifications jointly and clustering inference at the respondent level.

I then hold the estimation sample fixed and compare the family-structure estimate before and after adding the noncognitive measures. The first comparison adds conscientiousness and agreeableness within the Goldberg sample. The second additionally includes behavioral and emotional problems and restricts both the baseline and augmented specifications to the 4016 respondents for whom all three measures are observed. These same-sample comparison separate the change induced by adding the controls from the change induced by restricting the sample.

Results and interpretation Panel A of Table B.1 reports sample-sensitivity comparison. In the main sample, the estimated average marginal effect of single-parent household is 0.61 percentage points and is statistically significant at the 5 percent level. In the Goldberg sample, the estimate declines to 0.41 percentage points and is less precisely estimated. The difference between the two estimates is 0.20 percentage points, but the equality test does not reject the null that the two AMEs

Table B.1: Observed Noncognitive Characteristics and Post-School Unemployment

	Sample comparison		Same-sample comparison	
	Main	Goldberg	Baseline	+ C/A/ Behavior
<i>Panel A. Family-structure estimate</i>				
SP-TP difference in unemployment probability (percentage points)	0.61**	0.41	0.42	0.35
	(0.29)	(0.37)	(0.37)	(0.36)
Difference in AMEs (percentage points)		0.20		0.07*
Difference test, <i>p</i> -value		0.476		0.073
Attenuation		–		16.2%
<i>Panel B. Noncognitive characteristics</i>				
Conscientiousness	–	–	–	-0.48** (0.20)
Agreeableness	–	–	–	-0.00 (0.19)
Behavioral/emotional problems	–	–	–	0.30* (0.17)
Joint test of noncognitive controls, <i>p</i> -value	–	–	–	0.010
Respondents	7,347	4,029	4,016	4,016
Respondent-weeks	4,821,227	2,330,505	2,323,628	2,323,628

Notes: The dependent variable equals 1 if the respondent is unemployed and 0 if employed in a given post-school labor-force week. Entries for the single-parent indicator and noncognitive characteristics report average discrete changes in predicted unemployment probabilities from binary logit models and are expressed in percentage points. The main and Goldberg estimates in the first two columns are evaluated on their respective estimation samples. The final two columns use the identical sample of respondents with complete information on conscientiousness, agreeableness, and behavioral or emotional problems. The reported differences equal the baseline AME minus the corresponding comparison AME. The first difference compares the main and Goldberg samples. The second compares the baseline and noncognitive-control specifications estimated on the same sample. Because the same respondents contribute to both estimates in each comparison, the difference tests account for their common information by estimating the relevant specifications jointly and clustering inference at the respondent level. The noncognitive measures are converted into z-scores with a mean of 0 and a standard deviation of 1. Their estimated effects therefore correspond to a one-standard-deviation increase in the relevant characteristic. All specifications include the controls from the fully controlled unemployment specification in the main text. Standard errors are clustered at the respondent level and reported in parentheses. The joint-test row reports the *p*-value for the null that the included noncognitive controls are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

are the same ($p = 0.470$). Thus, although the estimate becomes smaller in the Goldberg sample, the results do not establish that the family-structure association becomes weaker among respondents in the Goldberg subsample.

Panel B reports the same-sample comparisons. The noncognitive measures themselves contain meaningful information about unemployment risk. For example, a one-standard-deviation increase in conscientiousness is associated with an approximately 0.48 percentage-point lower

probability of unemployment and is statistically significant at the 5 percent level. The estimate for agreeableness is close to zero and statistically insignificant. A one-standard-deviation increase in behavioral and emotional problems is associated with an approximately 0.30 percentage-point higher unemployment probability and is statistically significant at the 10 percent level. The controls are jointly significant: the joint-test p -value is 0.028 for conscientiousness and agreeableness and 0.010 when behavioral and emotional problems are included.

Adding these controls reduce the estimated family-structure association. Within the Goldberg sample, adding conscientiousness and agreeableness reduces the single-parent AME from 0.41 to 0.37 percentage points, an attenuation of approximately 9 percent. In the common sample with all three measures, the estimate declines from 0.42 to 0.35 percentage points. This decline of 0.07 percentage points corresponds to an attenuation of approximately 16 percent. The attenuation is statistically significant at the 10 percent level ($p = 0.073$). This provides suggestive evidence that observed noncognitive characteristics account for part of the family-structure difference in unemployment.

These results are informative but should be interpreted with caution. The family-structure estimate is already imprecisely estimated in the restricted-sample baseline, *before* the noncognitive controls are added. The comparison therefore does not show that adding these characteristics causes an otherwise precisely estimated association to disappear. Instead, it shows that conditioning on the observed noncognitive measures reduces the point estimate, with statistical evidence that is suggestive rather than conclusive.

The exercise has two additional limitations. First, the available NLSY97 measures capture only selected dimensions of noncognitive characteristics and cannot rule out other forms of persistent worker heterogeneity. Second, these characteristics may themselves partly reflect earlier family environments. Conditioning on them therefore does not identify a causal mediation effect of noncognitive characteristics.

I interpret this exercise as suggestive evidence that noncognitive characteristics may account for a limited but non-negligible portion of the association between family structure and unemployment. The measures themselves predict unemployment, and their inclusion reduces the estimated family-structure difference by approximately 16 percent. However, the available data do

not allow me to determine precisely how much of the gap is attributable to these characteristics or whether they represent an underlying mechanism. A more definitive analysis would require richer measure of noncognitive development observed at multiple stages of childhood and adolescence, together with information on family structures before these characteristics are formed. I therefore leave a more detailed investigation of this channel for future research when such data become available.

Appendix C Family Support and Geographic Constraints

The main analysis shows that workers from single-parent households experience a higher employment-to-unemployment (EU) hazard even after conditioning on observed human capital, parental education, and family financial resources. The main text studies whether this instability is associated with the types of jobs workers enter and with subsequent match stability. This appendix considers a complementary worker-side explanation: employment continuity may also depend on workers' geographic ties to family members and on access to informal time and logistical support.

I examine two margins. First, I use parental proximity to ask whether the family-structure difference in EU risk is related to workers' geographic location relative to their parents. The interpretation of proximity is theoretically ambiguous. Nearby parents may provide transportation, emergency assistance, temporary housing, or other forms of practical support that help a worker remain employed. At the same time, living close to parents may reflect family-care obligations or geographic attachment that restrict mobility or increase demands on the workers' time. I therefore treat parental proximity as a measure of family geography rather than as a resource per se.

Second, I examine grandparent availability as a proxy for an additional source of informal family support. A grandparent who is present in the household may provide time or logistical assistance that partly offsets differences in the family resources associated with single-parent household. The grandparent exercise therefore asks whether SP-TP difference in EU risk is smaller when an additional family member is available.

The two margins are not assigned the same prior interpretation. Parental proximity can represent either access to practical assistance or geographic and family-care constraints, so I treat its

expected relationship with employment stability as ambiguous. For grandparent availability, by contrast, exiting U.S. evidence provides a stronger basis for a buffering hypothesis: grandparents continue to provide adult grandchildren with emotional, financial, and practical support, including assistance with chores and transportation (Huo et al., 2018).

C.1 Parental Proximity

Measure construction The NLSY97 reports the respondent's distance from the respondent's mother and father in multiple survey waves. To avoid conditioning on location choices made after labor market entry, I construct the primary proximity measures using the latest valid distance observation from a survey year strictly before the respondent's terminal school-exit year.

For each respondent, I first construct the distance to the nearest observed parent and group the measure into four categories: same household, 1–30 miles, 31–100 miles, and more than 100 miles. I also construct mother and father-specific distant measures. As an additional specification, among respondents for whom both parental distances are observed in the same pre-exit survey wave, I measure the number of parents residing within 30 miles. Missing parental-distance observations are not coded as distant parents.

Empirical design I incorporate these measures into the fully controlled EU hazard specification reported in Table 4. Because parental proximity is observed only for a subset of respondents, all comparisons are made within the same proximity-observed sample. I first estimate the baseline EU hazard model without the proximity measures and then re-estimate the same specification after adding the parental-distance categories.

The primary comparison is the SP–TP EU hazard ratio before and after conditioning on parental proximity. If observed family geography accounts for an important part of the higher EU risk associated with single-parent household, the estimated family-structure difference should become quantitatively smaller after the proximity measures are added. I also test whether the proximity categories jointly predict the EU hazard and examine the pattern of EU risk across distance categories.

The sign of the proximity relationship is informative about competing mechanisms, although

Table C.1: Parental Proximity and EU Risk

Proximity measure	Same-sample baseline SP/TP HR	With proximity controls SP/TP HR	Proximity controls joint p -value	SP interactions joint p -value
<i>Panel A. Family-structure hazard ratio before and after proximity controls</i>				
Nearest observed parent	1.150	1.157	0.283	0.935
Mother	1.128	1.136	0.392	—
Father	1.118	1.117	0.944	—
Number of parents within 30 miles	1.079	1.047	0.176	—
<i>Panel B. EU hazard across nearest-parent distance categories</i>				
Distance to nearest observed parent			EU hazard ratio	p -value
Same household			1.000	Reference
1–30 miles			0.896	0.085
31–100 miles			1.061	0.621
More than 100 miles			0.949	0.473

Notes: The dependent variable is the weekly transition from employment to unemployment. SP/TP HR denotes the hazard ratio comparing respondents from single-parent households with respondents from two-parent households. Panel A compares the family-structure hazard ratio before and after adding each proximity measure. Baseline and adjusted estimates are calculated on the same proximity-observed sample. The joint p -value tests the corresponding proximity categories. The interaction test allows the association between nearest-parent proximity and EU risk to differ by family structure. The last row of Panel A is restricted to respondents for whom distances to both parents are observed in the same pre-exit survey wave. Panel B reports hazard ratios relative to respondents living in the same household as their nearest observed parent. All specifications include the controls in the fully controlled EU hazard model in Table 4. Standard errors are clustered at the individual level.

not causally identifying. A lower EU hazard among workers living closer to their parents would be consistent with nearby family members providing practical support. A higher EU hazard among workers living closer to their parents could instead be consistent with stronger geographic attachment, family-care obligations, or other time demands associated with remaining near family.

As an additional heterogeneity exercise, I allow the relationship between parental proximity and the EU hazard to vary by family structure. This specification asks whether geographic proximity is differentially associated with employment stability for workers from single-parent households.

Results and interpretation Panel A and B of Table C.1 report the parental proximity results. Panel A examines whether observed family geography accounts for the higher EU hazard associated with a single-parent household. In the nearest-parent sample, adding the proximity cate-

gories changes the *SP/TP* hazard ratio from 1.150 to 1.157. The corresponding changes are similarly small when distance to the mother or father is considered separately: the hazard ratio moves from 1.128 to 1.136 in the mother-proximity sample and from 1.118 to 1.117 in the father-proximity sample. The proximity categories are not jointly significant in any of these specifications, with p -values of 0.283, 0.392, and 0.944, respectively.

The additional specifications in Panel A lead to the same conclusion. Allowing the nearest-parent relation to differ by family structure produces a joint interaction p -value of 0.935, providing no evidence that proximity is differently related to employment stability for workers from single-parent households. In the more restrictive sample for which distances to both parents are observed in the same pre-exit survey wave, adding the number of parents living within 30 miles reduces the *SP/TP* hazard ratio from 1.079 to 1.047. However, both estimates are statistically close to 1 and the proximity measure is not jointly significant ($p=0.176$). Given the restricted sample and the absence of similar attenuation in the other specifications, I do not interpret this change as evidence that parental proximity explains the broader family-structure difference in EU risk.

Panel B examines whether the pattern of EU risk across distance categories is more consistent with a family support or geographic constraint interpretation. Relative to respondents living in the same household as their nearest observed parent, the EU hazard ratio is 0.896 for those living 1–30 miles away, 1.061 for those living 31 miles away, and 0.949 for those living more than 100 miles away. The corresponding p -values are 0.085, 0.621, and 0.473. The somewhat lower hazard among workers living 1–30 miles away could be consistent with parental coresidence reflecting family care obligations or other constraints. This pattern does not persist at greater distance. EU risk neither rises monotonically with distance as a family support interpretation would suggest, nor declines monotonically with distance as a geographic constraint interpretation might suggest.

The proximity results therefore do not support either mechanism and do not account for the estimated *SP/TP* hazard ratio. This conclusion is necessarily limited by what geographic distance measures. Workers living at similar distance from their parents may differ substantially in the transportation, emergency assistance, temporary housing, or other support they receive. Conversely, proximity does not directly measure family care responsibilities, time demands, or restrictions on geographic mobility. The results show that the observed measures of family geog-

raphy do not account for the higher EU hazard. However, they do not imply that family support or family-related constraints are unimportant, because geographic proximity is only an imperfect proxy for either mechanism.

C.2 Grandparent Availability

Measure construction I construct measures of grandparent availability from the NLSY97 household roster. The measures indicate whether at least one grandparent is co-resident with the respondent and should therefore be interpreted as proxies for the availability of an additional family member rather than as measures of realized support.

I use two main timing definitions. The first identifies grandparent co-residence in the household roster observed closest to age 16, restricting the observation to ages 15–17. This measure captures the extended-family environment during adolescence. The second identifies grandparent co-residence in the latest household roster observed in a calendar year strictly before the respondent’s terminal school-exit year. This measure is intended to capture potential family support available closer to the transition into the labor market.

I also consider a more recent pre-exit measure that requires the household roster to be observed within one or two years before school exit. A measure based on the latest roster observed in or before the school-exit year is used only as a sensitivity check because the available roster information does not permit exact ordering of the interview and school exit within the same calendar year.

Empirical design To examine whether the availability of an additional family member is associated with greater employment stability, I augment fully controlled EU hazard specification with grandparent availability and its interaction with single-parent household. The specification is

$$\log \left[-\log \left(1 - h_{it}^{EU} \right) \right] = \alpha + \beta SP_i + \theta GP_i + \phi(SP_i \times GP_i) + X'_{it}\gamma,$$

where SP_i indicates single-parent household, G_i indicates grandparent availability, and X_{it} contains the same demographic, own education, parental education, family financial resources, calendar-

time, and employment duration controls as the fully controlled transition specification in Table 4.

The specification yields two complementary comparisons. First, I compare the SP/TP EU hazard ratio among respondents without an available grandparent with the corresponding hazard ratio among respondents with an available grandparent. A smaller family-structure difference when $GP_i = 1$ would be consistent with a buffering role for additional family support.

Second, I compare the EU hazard between respondents with and without an available grandparent separately within the two-parent and single-parent groups. This comparison asks whether grandparent availability is itself associated with employment stability and whether that association is stronger for workers from single-parent households. The interaction coefficient ϕ provides a formal test of whether the family-structure difference in EU risk varies with grandparent availability. I also report the corresponding contrasts in predicted weekly EU probabilities.

The interpretation of this exercise is deliberately limited. Grandparent co-residence does not reveal whether the grandparent actually provides rides, care, financial assistance, or other time transfers. Moreover, co-residence may itself arise in response to family need. The estimates therefore indicate whether observed grandparent availability is associated with the pattern of employment instability predicted by a family support mechanism, rather than identifying a causal grandparent effect.

Results and interpretation Table C.2 reports results using grandparent co-residence as an observable proxy for grandparent availability. Panel A compares SP/TP EU hazard ratio by grandparent co-residence. Among respondents without a co-resident grandparent, the estimated hazard ratio ranges from 1.100 to 1.114 across the four timing definitions, and all four estimates are statistically significant at the 1 percent level. Among respondents with a co-resident grandparent, the corresponding estimates range from 0.936 to 1.081 and none is statistically distinguishable from 1.

Although the estimates for the GP -present group are consistently closer to 1, the difference in statistical significance across the two family-structure groups does not establish that grandparent co-residence moderates the family-structure gap. The relevant comparison is given by interaction test. The corresponding p -values ranges from 0.216 to 0.917, so none of the specifications rejects equality of the SP/TP log hazard ratios across grandparent co-residence status. Inference is also

Table C.2: Grandparent Co-residence and Employment-to-Unemployment Hazards

Panel A. Family-structure difference by grandparent co-residence

Timing of grandparent measure	SP/TP HR GP absent	SP/TP HR GP present	Interaction <i>p</i> -value	Respondents	
				GP absent	GP present
Strictly before school exit	1.106*** (0.042)	0.985 (0.134)	0.403	7,050	274
Within one or two years before exit	1.108*** (0.043)	1.062 (0.141)	0.755	6,445	249
Closest roster at ages 15–17	1.100*** (0.039)	1.081 (0.178)	0.917	7,016	355
In or before school-exit year	1.114*** (0.041)	0.936 (0.128)	0.216	7,168	255

Panel B. Grandparent association within family-structure groups

Timing of grandparent measure	GP HR within TP	GP HR within SP
Strictly before school exit	1.124 (0.124)	1.001 (0.085)
Within one or two years before exit	1.039 (0.109)	0.996 (0.087)
Closest roster at ages 15–17	1.136 (0.129)	1.116 (0.129)
In or before school-exit year	1.128 (0.118)	0.948 (0.089)

Notes: Panel A reports the *SP/TP* employment-to-unemployment hazard ratio separately for respondents with-out and with a co-resident grandparent. Panel B reports the grandparent hazard ratio (*GP HR*), defined as the *GP-present/GP-absent* hazard ratio, separately within the two-parent and single-parent groups. The final two columns in Panel A report the numbers of unique respondents with and without a co-resident grandparent, respectively, in each timing-specific estimation sample; the same sample counts apply to the corresponding specification in Panel B. Parentheses contain delta-method standard errors of the reported hazard ratios. The interaction *p*-value tests whether the *SP/TP* log hazard ratio differs by grandparent co-residence and is the formal test of the buffering hypothesis. The final timing definition is treated as a sensitivity specification because the ordering of the household interview and school exit within the calendar year is unknown. All specifications include the full set of controls used in Table 4. Standard errors are clustered by respondent. Statistical significance for the null that the corresponding hazard ratio equals 1 is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

limited by the small size of the *GP-present* sample, which contains only 249 to 355 respondents, or 3.4 to 4.8 percent of the corresponding estimation sample. The corresponding standard errors are therefore substantially larger than those of *GP-absent* group. The point estimates are descriptively consistent with a buffering role, but the interaction tests do not detect a statistically significant difference in the *SP/TP* hazard ratio by grandparent co-residence status. Given the small *GP-present* sample and the resulting large standard errors, the evidence on whether grandparent co-residence buffers the family-structure gap is inconclusive.

Panel B examines the association between grandparent co-residence and EU risk within each

family-structure gap. The *GP*-present/*GP*-absent hazard ratio ranges from 1.039 to 1.136 among workers from two-parent households and from 0.948 to 1.116 among workers from single-parent households. None of these within-group estimates is statistically significant, and their directions do not form a consistent pattern across timing definitions.

Taken together, the results do not provide statistically detectable evidence that grandparent co-residence buffers the higher EU hazard among workers from single-parent households. However, this conclusion does not imply that assistance from grandparents is unimportant. Co-residence measures potential availability rather than realized help with transportation, caregiving, temporary housing, financial needs, or job transitions. Therefore, the findings should be interpreted as a limited test based on an observable proxy for potential family support, rather than as evidence against a broader role for grandparent assistance.

Appendix D Construction of the Mechanism Decomposition

This appendix describes how I assign each model-generated family gap to the unemployment-search quality wedge, α_U , and the employed-search stability wedge, γ_E . The procedure follows the Shalpey decomposition. It is useful because the model is nonlinear: the change produced by equalizing one wedge can depend on whether the other wedge has already been equalized.

For a given outcome, let Δ_{ab} denote the model-generated *SP-TP* difference, where $a \in \{0, 1\}$ indicates whether the α_U is active and $b \in \{0, 1\}$ indicates whether the γ_E is active. The four model configurations are

$$\begin{aligned}\Delta_{11} &= \Delta(\hat{\alpha}_U, \hat{\gamma}_E), & \Delta_{10} &= \Delta(\hat{\alpha}_U, 0), \\ \Delta_{01} &= \Delta(0, \hat{\gamma}_E), & \Delta_{00} &= \Delta(0, 0).\end{aligned}$$

Thus, Δ_{11} is the difference generated by the estimated model, Δ_{10} retains only the quality wedge, Δ_{01} retains only the stability wedge, and Δ_{00} removes both wedges. Because the two family groups are otherwise identical in the model, $\Delta_{00} = 0$ for the outcomes considered here.

The contribution of the unemployment-search quality wedge is

$$\phi_\alpha = \frac{1}{2} [(\Delta_{11} - \Delta_{01}) + (\Delta_{10} - \Delta_{00})].$$

The first term measures the contribution of α_U when γ_E remain active, while the second measures its contribution when γ_E has already been removed.

Similarly, the contribution of the employed-search stability wedge is

$$\phi_\gamma = \frac{1}{2} [(\Delta_{11} - \Delta_{10}) + (\Delta_{01} - \Delta_{00})].$$

The two contributions satisfy the adding-up condition

$$\phi_\alpha + \phi_\gamma = \Delta_{11} - \Delta_{00}.$$

To make the role of the equalization order transparent, the two paths can also be written as

$$\Delta_{11} \xrightarrow{\alpha_U=0} \Delta_{01} \xrightarrow{\gamma_E=0} \Delta_{00}, \quad \Delta_{11} \xrightarrow{\gamma_E=0} \Delta_{10} \xrightarrow{\alpha_U=0} \Delta_{00}.$$

For consistency with the Shapley formulas, an order-specific contribution is reported as the gap before equalization minus the gap after equalization. It is therefore the amount of gap attributed to the wedge at that step, not the after-minus-before change in the reported outcome.

Table D.1 reports both the gap remaining in each of the four model configurations and the marginal contribution obtained along each equalization path. The results illustrate why averaging over the two orders is useful. For example, equalizing α_U first raises the stationary unemployment gap from 1.030691 to 1.092277 percentage points, so its first-order contribution is -0.0061586 point. If γ_E has already been equalized, the corresponding contribution of α_U is -0.029679 point. Their average is -0.045632 point, the Shapley contribution reported in the main text. Conversely, the stability wedge contributes 1.092277 points when it is removed second and 1.060370 points when it is removed first.

I report each contribution as a percentage of the full model-generated difference:

$$S_\alpha = 100 \times \frac{\phi_\alpha}{\Delta_{11} - \Delta_{00}}, \quad S_\gamma = 100 \times \frac{\phi_\gamma}{\Delta_{11} - \Delta_{00}}.$$

A negative share indicates that a wedge offsets rather than widens the full difference. Conse-

Table D.1: Order-Specific Mechanism Decomposition

Outcome	SP-TP gap under each model configuration			
	Δ_{11} Full model	Δ_{01} $\alpha_U = 0$	Δ_{10} $\gamma_E = 0$	Δ_{00} Both zero
Weekly EU probability (pp)	0.023323	0.023286	-0.000058	0
Stationary unemployment rate (pp)	1.030691	1.092277	-0.029679	0
Quality-standardized EU log hazard ratio, direct job-to-job entry	0.151773	0.151836	-0.000063	0
Low-quality share, jobs entered from unemployment (pp)	3.136483	0.048176	3.213982	0
High-quality share, jobs entered from unemployment (pp)	-2.541336	0.410500	-2.796992	0
Low-quality share, stationary employment (pp)	2.169655	1.112598	0.967091	0
High-quality share, stationary employment (pp)	-1.918618	-0.840705	-0.980004	0

Outcome	Equalize α_U first		Equalize γ_E first	
	α_U first $\Delta_{11} - \Delta_{01}$	γ_E second $\Delta_{01} - \Delta_{00}$	γ_E first $\Delta_{11} - \Delta_{10}$	α_U second $\Delta_{10} - \Delta_{00}$
Weekly EU probability (pp)	0.000037	0.023286	0.023381	-0.000058
Stationary unemployment rate (pp)	-0.061586	1.092277	1.060370	-0.029679
Quality-standardized EU log hazard ratio, direct job-to-job entry	-0.000063	0.151836	0.151836	-0.000063
Low-quality share, jobs entered from unemployment (pp)	3.088307	0.048176	-0.077499	3.213982
High-quality share, jobs entered from unemployment (pp)	-2.951836	0.410500	0.255656	-2.796992
Low-quality share, stationary employment (pp)	1.057058	1.112598	1.202564	0.967091
High-quality share, stationary employment (pp)	-1.077912	-0.840705	-0.938614	-0.980004

Notes: Each entry is an SP-TP difference. Probability and share differences are in percentage points; the quality-standardized EU hazard difference for jobs reached through direct job-to-job transitions is on the log-hazard-ratio scale. In the lower panel, a contribution is the gap before equalizing the indicated wedge minus the gap afterward. Thus, a negative entry means that the wedge offsets the full model-generated gap at that step. The Shapley contribution of each wedge is the average of its “first” and “second” entries. Small discrepancies can arise from rounding.

quently, when one wedge has a negative contribution, the contribution of the other wedge can exceed 100 percent while the two shares contribute to sum to 100 percent.

The scale of Δ_{ab} follows the outcome being decomposed. For stationary unemployment and aggregate weekly EU transitions, it is the SP-TP probability difference. For job-quality shares, it is the SP-TP difference in percentage points. For the job-quality-standardized employment-exit hazard ratio among jobs reached through direct job-to-job transitions, the decomposition is per-

formed on the log-hazard-ratio scale so that contributions are additive. All calculations are evaluated at the estimated parameter vector. They provide model accounting and do not incorporate parameter uncertainty or identify the causal effect of a particular policy intervention.

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